Two Arrows

Why a Learning Mind Heals Its Answers Long Before Its Self

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What this paper found

A mind has two arrows. The outer arrow is what it sees and does: its answers. The inner arrow is who it is: the inside that produces them. We sent a small learning network round closed loops of tasks, brought it back exactly to where it started, and measured both arrows on the way home.

1. **The answers come home; the inside does not, for a long time.** After a loop, the network practises its old task. After 30,000 practice steps its answers keep 0.0167 of the change they had on return, less than two percent, while its hidden layer still keeps 0.3049 of its mark: the inside is 18.49 times better kept than the answers, on average over the five starts. A checker that grades only answers would call the network healed. We call what stays behind identity drift. The inside does heal, slowly: weight decay puts its softest directions on a clock of 200,000 steps (Act 4).

2. **A learner that only follows its own gradient picks up an area law.** The part of the leftover that flips when the loop is walked the other way doubles when the enclosed area doubles: log-log slopes of 0.9545 to 1.0092 against the area, in five of five starts, in the weights and in the hidden layer alike. Nobody told the network how to carry anything. This is the closed-loop form of a known order effect between two lessons, the step size squared times their Lie bracket, here measured inside a learner (Act 1).

3. **Most of the mark ignores direction.** The part that is the same whichever way the loop is walked is about four times the area part (0.059825 against 0.014719 for a circle), and the biggest mark of all encloses no area: straight out and straight back leaves 0.16007, eleven times the circle’s area part. Played forward and then straight backwards, a curriculum erases exactly the part that flips and leaves close to twice the part that does not, pointing along it with cosines 0.9976 to 0.9993 after practice: coming back to where you started is not the same as never having left (Act 2).

4. **Identity drift comes in two kinds.** Besides the gradual mark there is a discrete one. On a task that does not care which way its direction points, half a turn closes the loop, and the network comes home with a pair of its hidden units exchanged. The answers are home, no amount of practice moves it back, and renaming the units removes 68 to 85 percent of its distance from home. The exchange is one event: nothing is exchanged through the first four eighths of the half turn in any start, and then a pair crosses, just past a quarter turn (Act 3).

5. **What stays can be forecast the moment the loop ends.** Healing is plain linear relaxation: every direction of the weights relaxes at the rate its own stiffness sets, and the size of the mark follows that forecast within 9.10 percent from 300 to 30,000 practice steps. The same algebra, with the practice labels added and no practice run, forecasts where the inside ends up after 30,000 steps: within 10 percent in 28 of 45 cases and within 25 percent in 37, median ratio 1.003. Re-linearised along the way, it brings the two hardest kinds of label change within 25 percent in 9 of 10 cases (Act 4).

6. **Only a kept record of the self brings it home.** Practising the true answers leaves 0.3049 of the inner mark. Adding a gentle pull toward a record of the hidden layer as it was before the loop leaves 0.0003, about a thousand times less. Copies of the drifted self do not help: a checker that grades the network with its own answers walks the inside 4.68 times further from home than the loop did, and an honest copy of the inside taken on return keeps the drift. Another network’s answers rewrite the inside even when they are nearly right: a twin raised on the same lessons, whose answers sit only 0.006931 to 0.012433 from the truth, moves the hidden layer 1.3053 to 2.2596 times as far as the loop did, and a stranger’s push grows in proportion to how far its answers sit from the truth (148 to 226 per unit). Home is the exact answers the network was raised on. A record that needs no view of the inside works too: home’s answers recorded on inputs two and a half times wider than any it trained on cut the inner mark to 0.0731 to 0.1259, about a third of plain practice’s, and to 0.0779 to 0.1019 on strange ground where nothing was recorded (Act 5).

7. **Hold on after the trip, not during it.** A learner that holds its inner record all through the journey follows the moving task worse, 1.382 to 2.156 times the loss of one that does not. Switched on only after the journey, the same record leaves 0.00019 to 0.00085 of the mark on unfamiliar inputs, 3.1 to 65.0 times better than keeping it on throughout, and costs the journey nothing. A purpose of the network's own does not hold the self: a drive toward an assigned target keeps 0.88 of the undriven mark, and a drive to give again its own remembered answers keeps 0.33 (Act 5). This is the known trade-off between stability and plasticity, given a number.
8. **One line from the matriarch's map holds a lineage.** Down generations that each walk the same trip and hand the network on, identity drift with no note grows 3.1994 to 4.5697 times in six generations. A note of one direction, 97 numbers and one target, read from the first generation's map holds it at 0.0314 to 0.0598 of that. Written at the door, right after the trip, the same line holds twelve generations level, largest over smallest 1.02 to 1.06; measured from the chain instead of the matriarch, it leaks, ending 4.97 to 7.83 times higher. A parent's note handed to its child works as well as the child's own, 0.982 to 1.010 times its drift at generation 12, and amended at bedtime with the parent's own change it arrives without a settling ring, within 2.84 percent at every generation (Act 6).
9. **Graded against the truth, identity drift hides; graded against a kept copy of home, it is loud.** A healed network's answers differ from home's by 1 to 2 percent of home's own error, and the outside listener that heard a mind's errors in our first paper feels them within 0.14 to 1.06 percent. Off home ground the answer change grows 9.07 to 14.46 times, but the network's own error grows 23.96 to 59.29 times, so against the truth the drift hides better with distance. Against a kept copy of home it does the opposite. In a run registered and carried out by the first paper's session, the healed answers sit 0.043 percent of the output scale from the copy's at home, 8.888 to 9.876 times further on inputs three times wider, and a new lesson learned by both makes the gap 2.801 to 9.356 times larger. A lesson at right angles to home hears the mark 10.58 to 129.6 times louder than home does. What the copy hears is softness: an even random change matched at home grows only 1.01 to 2.085 times, while a random change in the soft directions is about as loud as the loop's mark, 0.5965 to 1.005 of it (Act 7). Errors need an outside ear; identity needs a kept record.
10. **The picture, first.** In a hand-built agent with a slow inner arrow and a fast outer one, the inner arrow keeps no permanent turn in a flat world and comes back turned by the enclosed area on a curved one, as parallel transport must (Gauss-Bonnet, the Foucault pendulum, Berry's and Hannay's phases). A real outside anchor holds its drift to 3.106 degrees after twelve laps, while the honest internal copy leaves the full turn, 146.884 degrees against 147.070 with no anchor at all. That geometry is old; the agent reproduces it, and it is the picture the rest of the paper fills in with a learner (Act 0). Act 8 carries every result to the knob an engineer would turn.

The count. Before their runs we wrote down 164 predictions. 81 held and 83 missed, and every miss was given a redesign check; of the first studies' misses, 11 were re-checked after a redesign and 4 then held, while later redesigns are counted inside their batches. A run registered and carried out independently, in the first paper's session, held 6 of 6. Every verdict is in Appendix A, and the misses that changed the story are in Appendix B. A full rerun of every script in a fresh copy reproduces every data file and every figure byte for byte.

The two arrows

Every agent in this paper carries two arrows. The *outer arrow* is what it sees and does: its answers and its behaviour, the part of it the world can check. The *inner arrow* is the self as a whole: its identity and its direction, its motivation and its drive, and the memories it chooses to keep. When the world takes an agent round a closed loop and brings it back, the outer arrow can come home while the inner arrow does not. That gap is what we call *identity drift*.

A small learning network carries only part of an inner arrow, and the paper says which part each test can see. Its hidden layer, the way it represents the world it was taught, is our window on its identity, and most of the measurements look there. Its direction is the heading its own gradient gives it, step by step. The memory it chooses to keep appears when we give it something to keep: a record of itself taken before a journey (Act 5), or a single line written at the door after each trip and handed down a lineage (Act 6). Here "memory" means what is deliberately kept, not the weights themselves, which change with every lesson. A drive of its own, a purpose it pursues whatever the world asks, appears when a learner is given one (Act 5). Motivation in the full sense, wanting something, is beyond these learners, and we do not claim it for them.

Why two arrows. Two arrows let a small mind bend and drift with its world without losing itself. Picture life as a maze. The outer arrow does the walking: the turns and the steps, the timing of events, the quick response to danger. The inner arrow holds the map, the motivation and the self, and the outer arrow buffers and protects it. In the language of control, the inner arrow is the slower loop of a two-loop controller, with the longer hysteresis tail: a longer memory of where it has been. It changes little from step to step, and it should: an inner arrow that swung with every step would be no compass at all. Yet it can move when its own teacher speaks, the way an inner voice can, or the matriarch of a pod. The learners in this paper show both halves of that picture: their insides heal slowly on their own (Act 4), and come almost all the way home when a record of their own earlier self is held up to them (Act 5).

Loop walking. The paper reads drift with a method we call loop walking. Take something round a closed loop, bring it back exactly to where it started, and read what it carries home that the loop itself put there. That leftover is called holonomy. A Foucault pendulum is the classic case: it swings all day over the same floor, and the slow turning of its swing reads the turning of the Earth without anyone looking at the sky. Holonomy reads which way things turned more than how hard anything pushed. It is often subtle, and it often rides free, a side channel carried alongside the work the system was doing anyway. One way to learn about a monster in the water is to be eaten by it. Loop walking watches instead for the outline of a fin and the shape of a wake, and steers clear. Here the monster is identity drift, which no wrong answer announces, and the wake is what a learning network carries home inside itself after its answers have already come home.

Act 0: The picture, in a hand-built agent

The question. Before any learning, what does it mean for an agent with two arrows to drift, and what can hold it?

The agent. A hand-built agent carries two arrows. The outer arrow is fast: each step it closes half the gap to what the agent sees. The inner arrow is slow: on a circle it follows the outer arrow at 2 percent of the gap per step. What the agent does is a blend of the two, set by a mixing angle psi: the direction of cos(psi) times the outer arrow plus sin(psi) times the inner arrow. In words: psi = 0 means act on the outer arrow alone, psi = 90 degrees means act on the inner arrow alone, and 45 degrees means equal weight. Everything in this act is known physics or engineering; the agent only reproduces it, and it is the picture the learner of Acts 1 to 7 fills in (run IDs q1 to q5 name the prototype's output files).

It does the task, and the blend earns its place only when the senses are noisy. Tracking a moving target, the equal blend scores 0.9858 against 0.0019 for random actions. With clean senses the outer arrow alone does best (0.9993). With noisy senses (40 degrees of noise) the blend beats it, 0.9636 against 0.9200. The two arrows really are fast and slow: after a 90-degree jump the outer arrow is back within 10 degrees in 4 steps and the inner arrow takes 110, 27.5 times longer (run q1).

A flat world heals; a curved world keeps the area. Carry an arrow round a closed loop on a circle, and it comes back unchanged: a rigid partner's largest drift is 2.226e-14 degrees, and a slow follower's drift is lag that heals (0.002334 of it left after rest). Carry the inner arrow round a loop on a globe with a "don't twist" rule, the smallest change that keeps it at right angles to the outer arrow, which is the rule a Foucault pendulum obeys, and it comes back turned by the area the loop enclosed: the drift equals the enclosed solid angle within 1.4023 degrees, and within 0.036779 degrees when the exact smallest rotation is used. A triangle with three right-angle corners turns it by 90.1333 degrees, and laps add (run q2, q2r). What the agent does turns less than who it is. With equal weight on the two arrows, the action comes home turned 68.5363 degrees against the inner arrow's 105.5558, exactly what mixing a true arrow half and half with a turned one must give: the cosine of the action's turn is half of one plus the cosine of the inner arrow's turn (a reading after the data, run q2). The outer arrow follows the world, so it comes home true and hides part of the turn. This is the geometric phase of parallel transport (Gauss-Bonnet, Foucault, Berry and Hannay); it is the kind of law Act 1 finds in a learner's own weights.

Only an outside reference bounds the drift. After twelve laps on the globe, an agent with no anchor has drifted 147.070 degrees. One that checks against a real outside reference has drifted 3.106. Two fake anchors that can never say no, a yes-man and a mirror, leave the drift exactly as it was (difference 0.00e+00), and an honest internal copy of the inner arrow, carried by the same rule, drifts 146.884 degrees while reporting less than one degree of error on average (run q3). The copy is honest about everything but the turn. It stands between 143.7773 and 143.7787 degrees from the real anchor in every start, and its spread across starts is the real anchor's, 1.541 degrees against 32.982 with no anchor: it removes every stumble exactly as well as the outside reference does and misses only the turn it shares with the inner arrow (a reading after the data, run q3). This is the logic of inertial navigation: a second identical gyro cannot see the drift both gyros share. Act 5 finds the same three things in a learner: a record kept from outside the journey holds the self, a checker that cannot say no is worse than none, and an honest copy of the drifted self keeps the drift.

The best mix moves with the noise, and after a crisis the slow arrow should be subtracted. The best mixing angle is -1, 12, 41 and 63 degrees for sensor noise of 0, 20, 40 and 60 degrees, and 85 degrees once the inner arrow's speed is tuned (run q4): the noisier the senses, the more the agent should lean on its slow arrow. Right after a sudden jump in the world the opposite holds: after one round of thinking the best mix is -42 degrees, which subtracts the slow arrow, and scores 0.7775 against 0.4426 for the outer arrow alone (run q5r). The difference between a fast and a slow arrow points where the world is going, the lead compensator of control engineering.

What Act 0 shows. Drift is holonomy (Figure 1): an arrow carried round a loop comes back turned by what the loop enclosed when the space is curved, and comes back clean when it is flat. No check carried along with the drifting arrow can see the turn; only a reference from outside the journey can. The rest of the paper asks whether a learner that nobody built to carry anything shows the same geometry, and finds that it does: in the area law of its weights, in the soft mark that practice cannot heal, and in the records that bring it home.

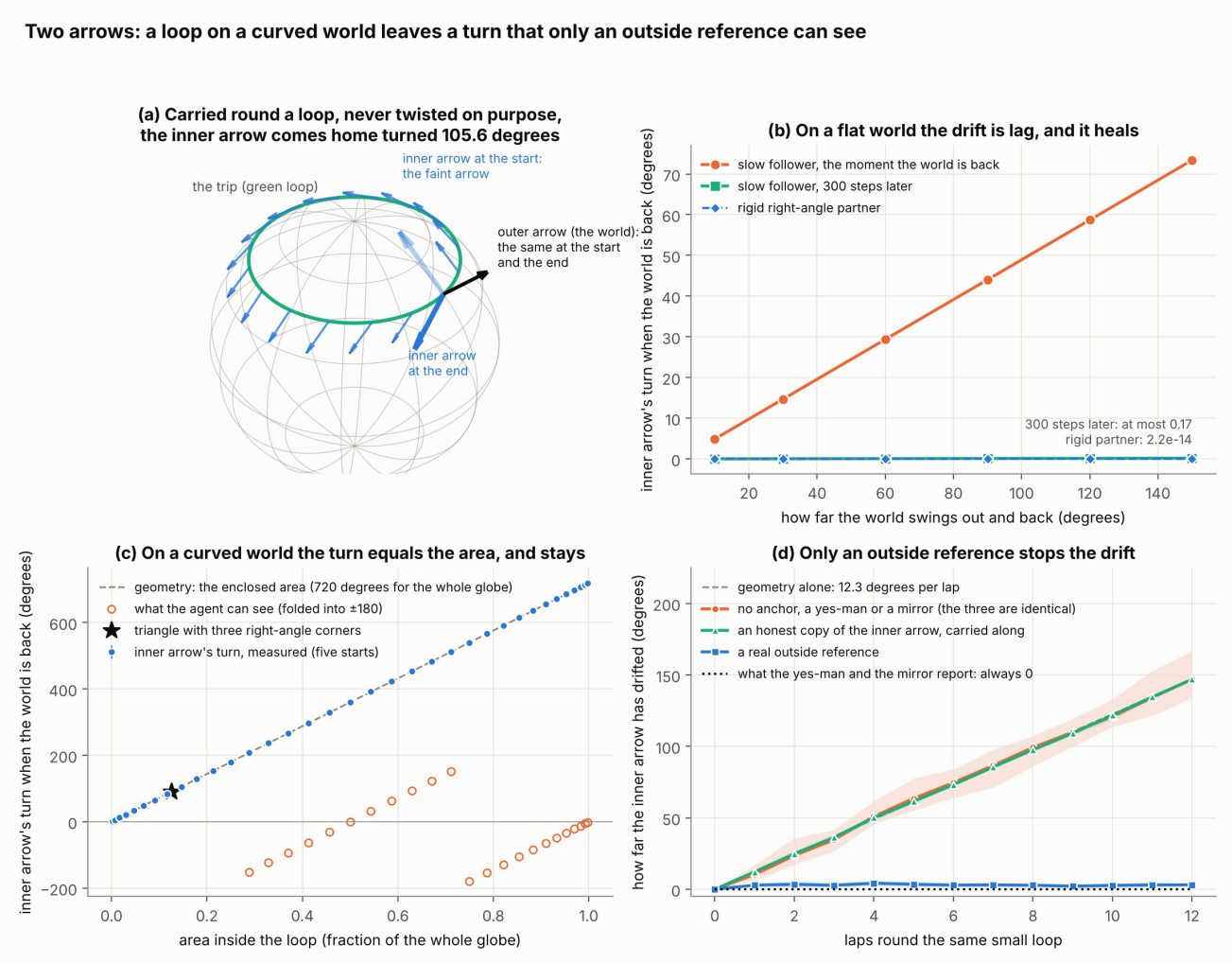


Figure 1. The hand-built agent with two arrows. (a) The inner arrow carried round a loop on a globe by the "don't twist" rule, the rule a Foucault pendulum obeys. The world, and with it the outer arrow, ends exactly where it began; the inner arrow comes home turned 105.6 degrees, the area of the cap it circled (105.442 degrees of solid angle), and the agent's trend arrow, mixed half and half from the two, turned 68.5 degrees (run q2). (b) On a flat world, a circle, a slow follower lags while the world swings out and back, and the lag heals: at most 0.17 degrees remain 300 steps later. A rigid partner never drifts at all (largest 2.226e-14 degrees; run q2). (c) On the globe the turn equals the area inside the loop, for every loop size within 1.4023 degrees, and a triangle with three right-angle corners turns it 90.1333 degrees; the open circles are the same turn folded into the plus or minus 180 degrees that the agent can see (run q2). (d) Twelve laps round a small loop. With no anchor, a yes-man or a mirror, the drift grows by 12.3 degrees a lap to 147.070 degrees, while the yes-man and the mirror report zero throughout; an honest copy carried along drifts with it (146.884); only a real outside reference holds it, at 3.106 (run q3). Lines are means over five starts; bands and bars run from the smallest to the largest (script figs/make_fig1_two_arrows.py).

Act 1: A learner picks up an area law

The question. If a learner keeps learning while its task wanders round a closed loop and comes back exactly to where it started, does the learner come back the same? And if it does not, does what it carries home follow a law?

The learner and the loop. The learner is a small neural network: 4 inputs, one hidden layer of 16 units that each squash their input with a tanh curve, and 1 output, 97 weights in all. Its task is to answer a smooth wave, y = sin(1.5 x·u) + 0.3 b, where x is the input, u is a direction in input space set by an angle a, and b both tilts that direction and shifts the answer up or down. In words: the task asks the network to read the input along one direction and answer with a wave, and the two knobs a and b choose the direction and the offset. Every pair of knob settings is a different task, so the two knobs make a flat map of tasks, the *task plane*. The network is first trained until it has learned the task at the centre of that map, a = b = 0. That is home.

Then the task moves. It travels round a circle of radius r in the task plane, centred on home, in 250 small moves, and at each move the network takes 4 small steps of plain gradient descent (step size 0.05), 1,000 steps per lap. Nobody tells the network how to carry anything; it only follows its own gradient, as any learner does when its world changes under it. When the task is back at home, the loop is closed. A second network, the *control*, starts from the same weights and trains on the home task for the same 1,000 steps. The difference between the two is the *leftover*, the mark the loop left behind. We follow the leftover in two places: in the weights, and in the hidden layer, whose activity on a fixed set of 500 probe inputs is our window on the learner's inner arrow: its identity, the way it represents its world. The output, what the network answers, is its outer arrow.

Two tricks to read the mark. The raw leftover is dominated by lag. A learner that follows a moving target always runs a little behind it, and that lag grows in step with the size of the loop. Lag is not what we are after, and two tricks remove it.

The first trick is the *turned copy*. The same loop is run a second time, rotated half a turn about home, and the two leftovers are averaged. Lag points the opposite way on the two copies and cancels, while anything that depends on the loop's shape rather than its position survives. It is a two-element phased array: two emitters half a turn apart cancel every odd-order term and keep every even one. The second trick is *time reversal*. The loop is driven forward and then, from a fresh start, backwards. Half the difference of the two leftovers is the part that flips sign with direction; call it A. Half their sum is the part that stays the same both ways; call it S. In words: A is what the loop's sense of rotation wrote, and S is what simply walking the loop wrote.

What theory predicts. Two lessons taken in opposite orders leave a learner in slightly different places, and the difference is the step size squared times a quantity called the Lie bracket of the two lessons' gradients (Rukhovich, Podolskiy and Piontkovskaya 2025; Dherin and colleagues 2025; the related implicit-regularization analysis of Smith, Dherin, Barrett and De 2021). Add up that order effect all the way round a closed loop and, by Stokes' theorem, the total is proportional to the signed area the loop encloses. In words: going round one way and coming back leaves a turn whose size is set by the area inside the loop, not by its length. So A should double when the area doubles, and since the area of a circle is pi r squared, A plotted against r squared on log-log axes should have a slope of 1. This is the same geometry as parallel transport on a curved surface, the Foucault pendulum and Berry's phase (Act 0); here the "curvature" is the learner's own.

What we found. The part that flips with direction follows the area law (Figure 2). Across circles of radius 0.025, 0.05, 0.1 and 0.2, the log-log slope of the size of A against r squared is 0.9545 to 1.0092 in the weights, five starts of five (run s1, P1), and 0.9503 to 1.0067 in the hidden layer (P3). It survives practice: after 3,000 steps of practising the home task, the slopes are still 0.8886 to 1.0092 in the weights and 0.8847 to 1.0079 in the hidden layer (P2, P3). It points the same way at every size: the cosine between A at each smaller radius and A at r = 0.2 is at least 0.9379 (P4). And its size per unit of area is close to a constant, about 0.12 weight units per unit of enclosed area at return for every radius from 0.025 to 0.4 (0.11713 to 0.12769).

The tricks matter. Without the turned copy, the same measurement gives slopes of 0.4931 to 0.5241 (P5): half the area law, because lag grows with r, and r is the square root of r squared. Lag swamps the mark until the turned copy cancels it.

Two honest limits on the headline. First, at return about half of A is not area. A figure-eight loop shaped like an L, which encloses no net area, still leaves 0.47 to 0.52 of the circle's A at return, whatever its size. That half comes from the order in which the loop visits places, and it heals fast: after 3,000 practice steps the L-figure-eight keeps 0.11 to 0.17 of the circle's A (run r2). What persists is mostly area. Second, A is not the biggest part of the mark. The part that ignores direction, S, is about four times larger, and a loop that goes out and back along a straight line, enclosing nothing, leaves the biggest mark of all. That larger mark is Act 2.

Speed and laps, the knobs. How fast the loop is walked sets how much area law it leaves, and there is a best speed. At return, A grows from 0.0048569 at 256 steps per lap to 0.017886 at 4,000, then falls back to 0.015457 at 16,000 (runs s3, r1). A fast loop cuts corners, the way a fast outer arrow cut inside a loop it could not follow in the hand-built agent of Act 0. A very slow loop lets more of the network's directions keep up with the task, and whatever keeps up does not keep a mark; in the limit of an infinitely slow loop the mark must vanish, because home is an isolated minimum. Laps add for the first few: after 3,000 practice steps, two laps leave 0.94289 to 1.0462 times twice the one-lap mark, pointing the same way (cosines at least 0.9952), and four laps 0.84928 to 1.1119 times four. By eight laps the mark wanders in size (0.5834 to 1.1827 times eight) and direction (cosines 0.8013 to 0.9771), because each lap is written from a network that the earlier laps have already moved.

What Act 1 shows. A learner that only follows its own gradient carries a geometric mark home from a closed loop of tasks, and the part of that mark that knows which way the loop was walked grows with the area the loop enclosed, in the weights and in the hidden layer alike. The algebra of the order effect between two lessons was known; the area law it implies has now been seen in a learner's own inside, with a measurement design (turned copy plus time

reversal) that separates it from lag. The knobs are the loop's area (the mark scales with it), its speed (a best speed near 2,000 to 4,000 steps per lap for this learner), the number of laps (they add for the first few), and the instrument itself: without the turned copy, the area law reads as half a law.

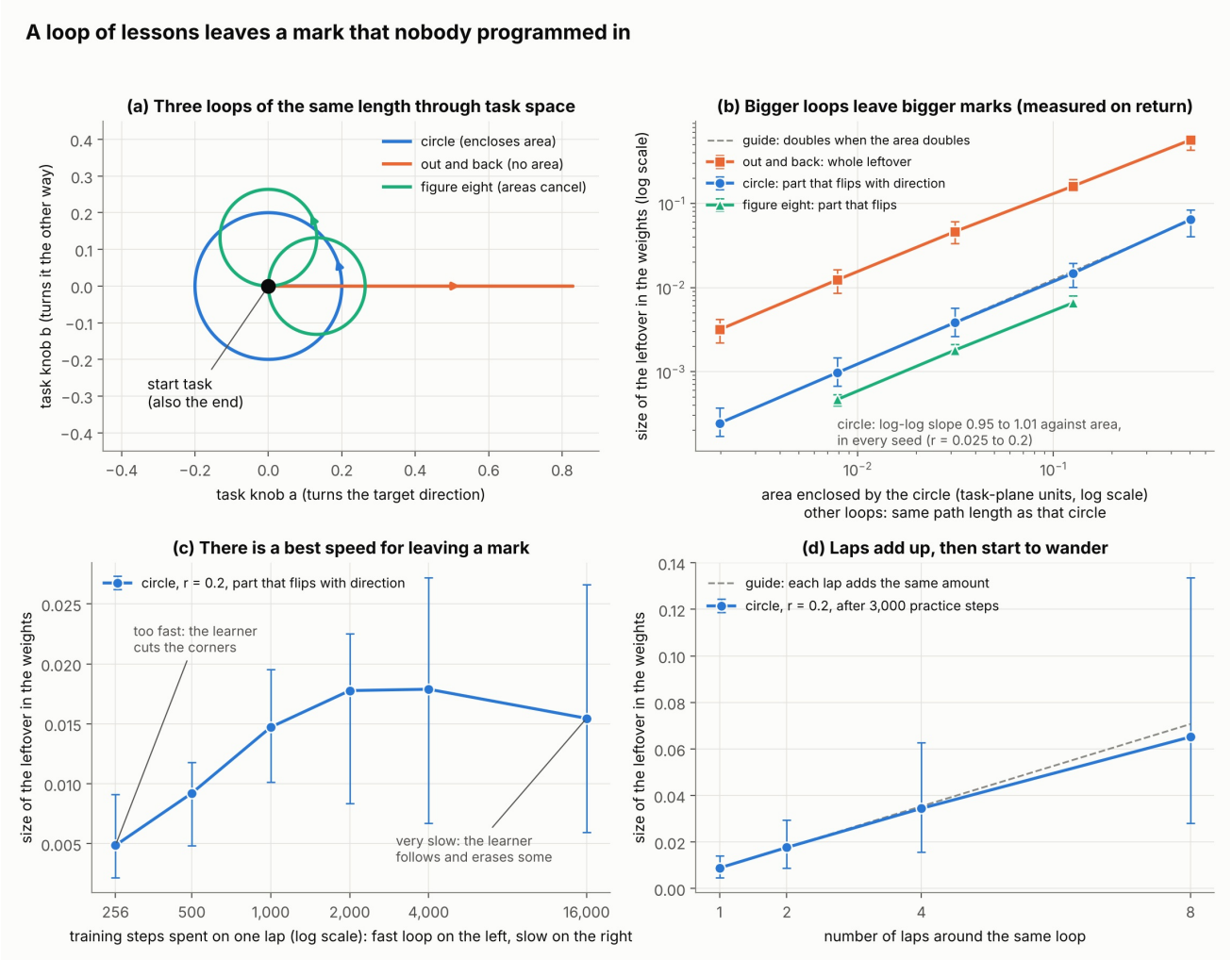


Figure 2. A loop of lessons leaves a mark that nobody programmed in. (a) Three loops of the same length in the task plane: a circle, which encloses area; a straight out-and-back, which encloses none; and an L-shaped figure eight, whose two lobes cancel. (b) The size of the leftover in the weights at return against the circle's area, on log-log axes: the circle's part that flips with direction runs along the guide line that doubles when the area doubles; the figure eight's flipping part sits below it; the out-and-back's whole leftover sits above both. (c) The best speed for leaving a mark. (d) Laps add up, then start to wander. Points are means over five starts; bars run from the smallest to the largest (runs s1, s3, r1, r2; script figs/make_fig2_area_law.py).

Act 2: The bigger mark that ignores direction

The question. The area law of Act 1 is the part of the mark that knows which way the loop was walked. What is the rest of it, the part that comes out the same whichever way the learner goes round?

It is the larger part. Recall the split of Act 1: A is half the difference between the loop walked forward and walked backwards, and S is half their sum, the part that is the same both ways. At return from a circle of radius 0.2, S is 0.059825 against A's 0.014719, about four times larger (run r1; Figure 3a). It grows with the square of the loop's size, just as A does: it is four times larger each time the radius doubles, from 0.0009351 at r = 0.025 to 0.05982 at r = 0.2 (run s1). And the loop that leaves the biggest mark of all encloses no area. Walk the task straight out along one knob and straight back, the same distance as the circle, and the leftover is 0.16007, eleven times the circle's area part (run r1; 7.052 to 15 times by start, s1). Going somewhere and coming back marks a learner more than going round. The straight excursion's mark bends a little below the square law as it grows: between neighbouring sizes its growth slope falls from 1.96 at the smallest loops to 1.79 between radius 0.1 and 0.2, while the circle's blind part holds at 2.00 (a reading after the data, run s1). Beyond radius 0.2 every part scatters from start to start, so whether the straight excursion levels off cannot be told from these runs.

It is not lag, and it is not the step size. The first guess was lag, the learner running behind a moving target, which should shrink as the loop slows down. It does not: walking the loop sixteen times more slowly, from 1,000 to 16,000 steps per lap, changes S with log-log slopes between -0.3004 and 0.1171, far from the slope of -1 that lag would give (run r1). The second guess was the finite size of the learner's steps, the effect behind the implicit regularization of gradient descent. Halving the step size while keeping the learning time the same changes A, S and the out-and-back leftover by less than half a percent (ratios 0.9966 to 1.0003, run f1). So the direction-blind mark belongs to the smooth flow of learning itself, not to any detail of how the steps are taken.

The pace leaves a permanent partial imprint. Walk the out-and-back fast on the way out and slowly on the way back (125 steps out, 500 back), then the other way round, and compare the two leftovers. Their difference, measured as rho, the size of their difference over the size of their sum, is 0.5447 to 0.6189 at return, and the two point in visibly different directions (cosines 0.5617 to 0.7182; run b1 T1). Their average sits between the leftovers of the all-fast and the all-slow walks in every start. Practice then heals most of the difference: rho falls to 0.0586 to 0.2778 after 3,000 practice steps. Then it stops falling. After 30,000 steps it is still 0.0661 to 0.3348 (run b2 R1). Most of the mark that persists has forgotten how fast the learner walked, and a part of it, different in size from start to start, remembers the pace for good. Because practice is plain linear relaxation (Act 4), that part lies in the soft directions that practice cannot reach.

Played backwards, a curriculum erases its order and doubles the rest. Walk the circle forward and then, with no pause, walk it backwards. The two halves together leave a mark that points along S (cosine 0.9520 to 0.9768 at return, 0.9976 to 0.9993 after 3,000 practice steps) and is close to twice S's size: twice S is missed by 0.2504 to 0.3387 of itself at return and by 0.0546 to 0.1191 after practice (run b9 E). In words: running the tape backwards removes exactly the part of the mark that depended on the order of the steps, the area part, and adds a second copy of the part that came from walking at all. The turned copy of Act 1 is the matching move in space, two emitters half a turn apart in a phased array; time reversal is the same move in time.

A reading, not yet a mechanism. A mark that grows with the square of the excursion, does not care how fast the excursion was made, and survives practice behaves like the remanence a small field excursion leaves in a magnetic material, which also grows with the square of the excursion (Rayleigh's law of hysteresis; Rayleigh 1887). We offer that only as a description of the behaviour. A forecast that uses only the home curvature and the path's pull, with no training, gets the blind mark's size to within 0.62 to 0.90 of the measured but its direction only partly (cosines 0.50 to 0.72; run b18): part of the blind mark is the learner's linear response to the path, and the rest comes from its own changing curvature along the way.

What Act 2 shows. Most of what a closed loop of lessons leaves in a learner does not depend on the direction of the loop, its speed or the size of the learner's steps. It grows with the square of the loop's size, it is largest for loops that enclose nothing, and practice heals part of it and leaves the rest. The knobs: the loop's size (the blind mark scales with its square), its shape (out-and-back marks most), its pace (an uneven pace leaves a partial imprint that never heals), and the order of play (walking it backwards erases the area part and doubles the blind part). For anyone designing a curriculum, the lesson is that coming back to where you started is not the same as never having left, and the biggest difference is made by going away and returning, not by going round.

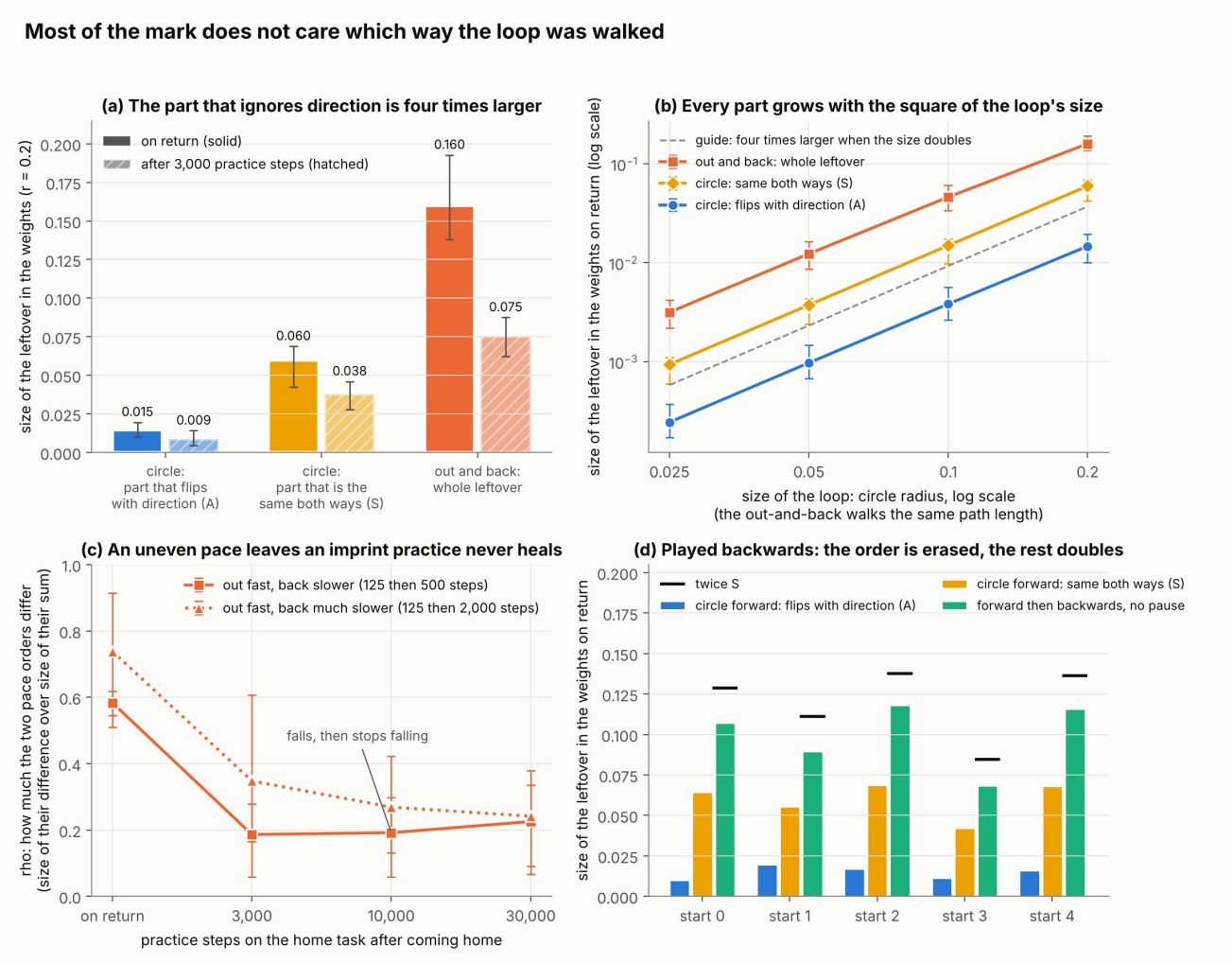


Figure 3. Most of the mark does not care which way the loop was walked. (a) The mark on return from a circle of radius 0.2, split into the part that flips with direction (A) and the part that is the same both ways (S), beside the whole leftover of an out-and-back walk of the same length; solid bars on return, hatched after 3,000 practice steps (run s1). (b) Every part grows with the square of the loop's size: by start, the log-log slopes against the radius are 1.9893 to 2.0464 for S, 1.9089 to 2.0184 for A and 1.8447 to 1.9887 for the out-and-back (run s1). (c) rho, how much the leftovers of the two orders of an uneven pace differ, as practice goes on. For the 125-then-500 walk it falls from 0.5447 to 0.6189 on return to 0.0586 to 0.2778 after 3,000 steps and then stops falling; the much slower return (dotted) is still 0.0917 to 0.3796 apart after 30,000 steps (runs b1 T1, b2 R1). (d) The circle walked forward and then straight backwards (green) leaves a mark that points along S and comes close to twice its size (black marks). On return it misses twice S by 0.2504 to 0.3387 of itself; after 3,000 practice steps by only 0.0546 to 0.1191, pointing along S with cosines 0.9976 to 0.9993 (run b9 E). Bars and points are means over five starts; whiskers run from the smallest to the largest (script figs/make_fig3_blind_mark.py).

Act 3: Two kinds of identity drift

The question. Acts 1 and 2 found a mark that grows by degrees. Can a learner also come home changed by a jump, all at once, with nothing to show for it in its answers?

A loop that flips the frame. Take a task that does not care which way its direction points: y = cos(1.5 x·u), where u is a direction in the plane of the first two inputs, turned by an angle a. In words: the network reads the input along one direction and answers with a wave that is the same whether the direction points one way or the opposite way, because the cosine is even. Now turn the direction half way round, from a = 0 to a = pi. The task at the end is exactly the task at the start (they differ by 1.5e-16 on the probe inputs), but the direction inside it has been reversed. A half turn is a closed loop of tasks that hands the learner back its own task with the frame flipped, the way a walk once round a Möbius strip brings you back to where you started, upside down.

The learner comes back with two of its parts exchanged. After the half turn and 30,000 steps of practice, the answers are home: the probe loss is 0.002894 against the control network's 0.002891 in the first start (run b4). Yet the network has not healed toward home at all. Its raw distance from home in weight space stays where the turn left it, 3.1758 to 3.6249 after 30,000 steps (run b5), and unit by unit its hidden layer reads like a stranger's (a relative hidden change of 0.8851 to 1.1422). The explanation is a relabelling. A network's hidden units can be listed in any order, and because tanh is an odd function a unit can also be flipped in sign with its outgoing weight, without changing anything the network computes; relabelling a network this way changes its answers by about 1e-16 (run b4). Matching each unit of the returned network to its best partner at home removes 68 to 85 percent of its distance from home (ratios 0.1499 to 0.3219), and in four starts of five exactly two units have traded places (run b5; Figure 4b). What remains after the relabelling is a hidden change of 0.1184 to 0.3696, the ordinary kind of mark. Practice cannot undo the trade, because the network with two units exchanged is an equally good answer to the same task: there is nothing for the gradient to push against.

Two controls show it is the flip, not the travel. A turn that goes a quarter of the way out and comes straight back has nothing to relabel (ratio 1.0000 in every start) and heals as usual. A whole turn, which brings the direction back the right way up, does not undo the half turn's trade either: it also ends with two units exchanged in four starts, and keeps more ordinary difference besides (ratios 0.2813 to 0.7997).

The trade is one event, at a quarter turn. Following the half turn in eighths and checking how much relabelling helps at each, nothing is relabelled through the first four eighths in any start; between a = pi/2 and a = 5pi/8 a pair of units crosses (relabelling now removes 0.183 to 0.237 of the distance), and the gain then grows to 0.601 to 0.795 by the end of the turn (run b6; Figure 4a). The crossing sits where the task's direction stands at right angles to where it began. A second half turn after practice undoes the trade in three starts of five, bringing the network back with nothing relabelled, and exchanges a different pair in the other two. The trades compose like moves in a larger reshuffling, not like one switch flipped twice.

Which two units trade? A unit's mirror partner is the unit that comes closest to its mirror image: the same unit with its weights on the first two inputs reversed, as the half turn reverses them (tanh's sign flip allowed). In three starts the two units that trade are each other's closest mirror partners ([6, 7], [1, 15] and [3, 15]; run b7). In the fourth, the four units that trade include a pair that are each other's partners ([10, 14]). In the fifth, unit 8's partner is unit 12, but unit 12's own partner is unit 4, and the trade exchanges [8, 12]; turning the other way exchanges [4, 12] instead. The half turn trades the units that its reflection makes most alike, where there are clean pairs to trade; with each group of clones merged into one unit first, the traded set is closed under mirror pairing in four starts of five (run b18).

Twins, and what is left over. The same relabelling lens says something about twins, networks trained on the same task from different random starts. In three starts of five, matching units removes 70 to 80 percent of the distance between a network and its twin (ratios 0.2011 to 0.3032): the twin is largely the same mind with its parts listed in another order. In the other two it removes only 36 to 42 percent (ratios 0.5838 and 0.6415): those twins are different minds that give the same answers (run b4). And the ordinary mark left after a half turn's trade ignores direction: turning forward or backward leaves leftovers that point the same way (cosines +0.1337 to +0.9871, run b6), the direction-blind kind of Act 2.

What Act 3 shows. Identity drift comes in two kinds. The gradual kind grows by degrees with the loop's size and shape (Acts 1 and 2). The discrete kind is a relabelling: the same parts, two of them exchanging their roles, in one event at a quarter turn, after which no amount of practice brings them back. The discrete kind is invisible to every check on answers by symmetry, since a relabelled network computes exactly the same thing; only a record that names the parts, a kept record of the inner self, can tell that it happened. The knobs: the loop's shape in task space (a half turn of a task that ignores direction flips the frame), the point of no return (a quarter turn), a second turn (which may undo the trade or make a new one), and the network's mirror pairs, which mark the units most likely to trade.

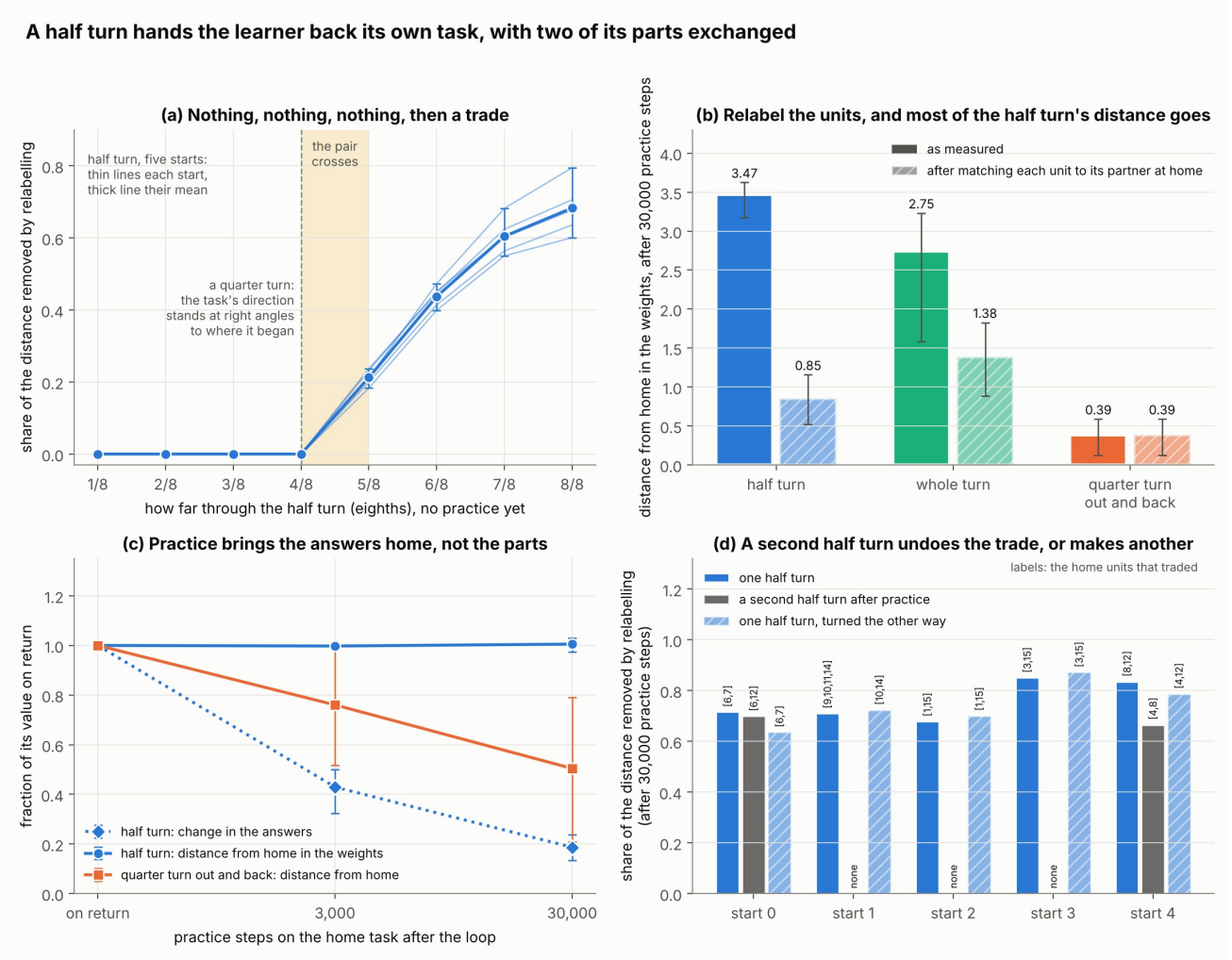


Figure 4. A half turn hands the learner back its own task, with two of its parts exchanged. (a) The share of the distance from home that relabelling the hidden units removes, after each eighth of the half turn and before any practice: zero through the first four eighths in every start, then 0.183 to 0.237 at five eighths, growing to 0.601 to 0.795 by the end (run b6). (b) Distance from home in the weights after 30,000 practice steps, as measured and after matching each unit to its partner at home: relabelling removes 68 to 85 percent of the half turn's distance, 20 to 72 percent of the whole turn's and none of the quarter turn out and back's (ratios 0.1499 to 0.3219, 0.2813 to 0.7997 and 1.0000; run b5). (c) Over practice the half-turned learner's answers come home, their change falling to 0.1331 to 0.2384 of its value on return, while its distance from home stays where the turn left it (0.9731 to 1.0302 of its value on return); the quarter turn out and back heals (0.2104 to 0.7917) (run b4). (d) The share removed by relabelling after one half turn, after a second half turn that follows practice, and after one half turn the other way, with the home units that traded written above each bar: the second half turn undoes the trade in three starts and trades a new pair in two (run b6). Bars and points are means over five starts; whiskers run from the smallest to the largest (script figs/make_fig4_half_turn.py).

Act 4: The answers come home; the inside does not

The question. After a closed loop of lessons, the learner goes back to practising its home task. Practice is the usual cure: whatever the loop disturbed, the home task's gradient pulls back. Does practice bring the whole learner home, or only the part of it that the home task can see?

Only the answers. Run the circle loop of radius 0.2 forward and as its turned copy (Act 1's trick, which removes lag), then practise the home task and watch two things: how far the learner's answers sit from the control's, and how far its hidden layer sits from the control's. After 3,000 practice steps the answers keep 0.0914 of the change they had on return, while the hidden layer keeps 0.675. After 30,000 steps the answers keep 0.0167, less than two percent, while the hidden layer still keeps 0.3049 of its mark (0.2453 to 0.3606 by start; run r3; Figure 5a). The inside is 8.208 times

better kept than the answers after 3,000 steps and 18.49 times after 30,000 (the average of the five starts' ratios, 14.99 to 22.19 by start). A checker that grades only answers would call this learner healed. Its outer arrow has come home; its inner arrow has not.

Why: practice is plain linear relaxation. The learner's weights near home can be described by their directions of curvature, the directions along which the home task's loss rises steeply (stiff) or hardly at all (soft). The home networks are very sloppy: 65 to 79 of their 97 weight directions have curvature below 0.001, while the stiffest reach 4.29 to 7.16 (runs s0, r1). Practice by gradient descent relaxes each direction on its own, at a rate set by its own curvature: after t steps a stiff direction keeps a fraction (1 - eta lambda)^t of its starting value, where eta is the step size and lambda the direction's curvature. In words: every direction shrinks by the same small factor at every step, fast for stiff directions and almost not at all for soft ones. That rule, with no free number in it, matches what happens. From 300 to 30,000 practice steps it gives the size of the leftover within 9.10 percent and the share of it lying in soft directions within 0.0442, in every start (run b1 T4; Figure 5b). The soft share itself tells the story: 0.0278 of the leftover lies in soft directions on return, and 0.9439 after 30,000 steps. Practice does not move the mark into hiding; it removes everything that is not already hidden. The answers depend mostly on stiff directions, so they heal; the hidden layer carries the soft part, so it keeps it. The inside is not marked for ever, only healed slowly: the small weight decay gives even the softest direction a curvature of at least 0.0001, so at this step size the softest directions heal on a clock of 200,000 steps (one over the step size times the weight decay) and still keep 0.8607 of themselves after 30,000.

So the outcome can be forecast, without running the practice. If practice is linear relaxation, then where it ends up can be written down the moment the loop ends. The same holds when the learner practises labels other than its own home answers, as long as the pull of the changed labels is added: each direction then relaxes from where it started toward a resting point equal to the pull along it divided by its curvature. In words: soft directions move far for a small pull, stiff ones hardly move at all. Across 45 cases, nine kinds of practice labels in five starts, this forecast lands within 10 percent of the measured hidden change after 30,000 steps in 28 cases and within 25 percent in 37, with a median ratio of 1.003 (run b7; Figure 5c). Its failures are the cases where the change of labels is large, or shaped unlike anything the network already does, where a first-order rule should be expected to fail. Recomputing the curvature and the pull ten times along the way, still without running any practice, rescues them: the two hardest kinds of label change land within 25 percent in 9 of 10 cases (run b18).

The clones never split, because of symmetry. Eight to eleven of the 16 hidden units in each home network are clones, identical copies of one feature. The loop's mark puts about a billionth of its squared size into the directions that would pull clones apart, where a random direction would put 0.4330 to 0.6186 (run r1). Gradient descent gives identical units identical pushes, so a loop can only move clones together. Break the symmetry at the start with a little weight noise and the share grows as the square of the noise (log-log slopes 1.9335 to 2.0064, every start; run b1 T5; Figure 5d), as a symmetry broken at first order should.

What Act 4 shows. Practice heals what a learner's answers depend on and leaves the rest, at rates that linear theory predicts exactly. The part that stays is the part in the soft directions, and its fate can be forecast at the moment the loop ends. The knobs: the length of practice (answers heal within a few thousand steps; the inside keeps about a third of its mark after 30,000), the stiffness spectrum of the home network (the more soft directions, the more is kept), and the labels practised, whose pull along each soft direction sets where the inside comes to rest. For anyone who checks a learner after a change, the warning is plain: healed answers say nothing about a healed inside. The next act asks what does bring the inside home.

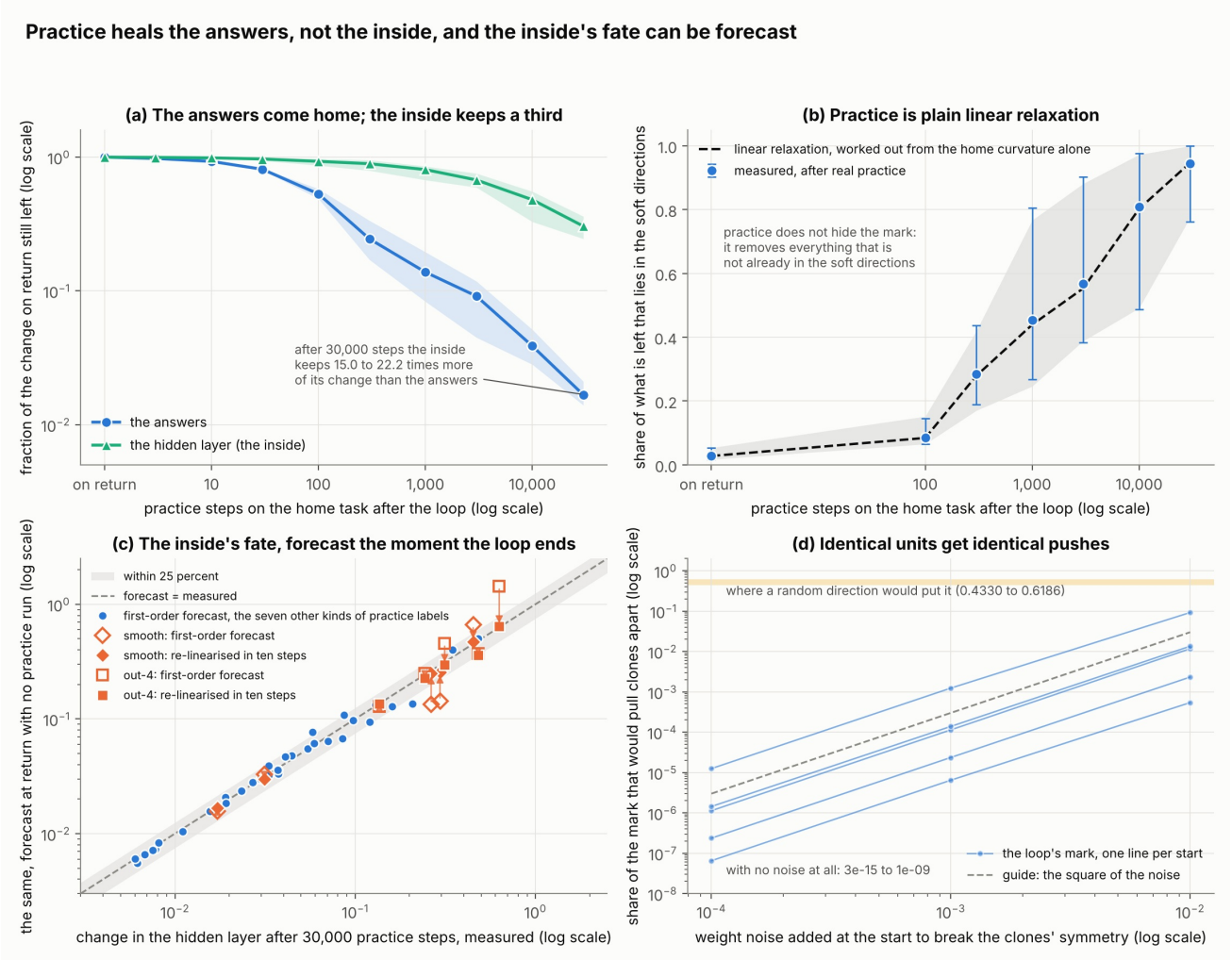


Figure 5. Practice heals the answers, not the inside, and the inside's fate can be forecast. (a) After a circle of radius 0.2 walked forward and as its turned copy, the share of the change on return still left in the answers and in the hidden layer as the learner practises its home task: after 30,000 steps the answers keep 0.0167 and the hidden layer 0.3049, which is 14.99 to 22.19 times more by start (run r3). (b) The share of what is left that lies in the soft directions, measured after real practice and worked out by linear relaxation from the home curvature alone; from 300 steps on, the two never differ by more than 0.0442 (run b1 T4). (c) The hidden change after 30,000 practice steps, forecast at return against measured, for 45 cases of nine kinds of practice labels: 28 within 10 percent and 37 within 25 percent, median ratio 1.003 (run b7). The open orange markers are the two kinds that fail at first order; the arrows show where the forecast lands when the curvature and the pull are recomputed ten times along the way, 0.861 to 1.045 and 0.746 to 1.021 of the measured (run b18). (d) The share of the loop's mark that would pull identical units apart, against weight noise added at the start to break their symmetry: about a billionth or less with no noise, growing as the square of the noise (slopes 1.9335 to 2.0064), and far below the 0.4330 to 0.6186 that a random direction would carry (runs b1 T5, r1). Lines are means over five starts and bands run from the smallest to the largest, except in (d), where each start is drawn (script figs/make_fig5_heal.py).

Act 5: What brings the inside home

The question. Practice on the true answers brings a learner's answers home and leaves a third of its inner mark behind (Act 4). Is there anything a learner can hold on to that brings the inside home as well? And what does holding on cost?

Eight things to hold on to. After the circle loop, each learner practised the home task for 30,000 steps while holding on to one of eight anchors, and we measured how much of the inner mark it kept (run b2; Figure 6a):

What the practice holds on to	Answers left	Inside left
the true answers	0.0167	0.3049
its own current answers (a checker that cannot say no)	10.8444	4.6840
its own answers recorded on return	1.2273	1.7621
its own hidden layer recorded on return	0.4934	0.9384
an independently trained network’s answers	0.9717	5.6064
the true answers and a record of its hidden layer from before the loop	0.0008	0.0003
the same record at a tenth of the strength	0.0017	0.0041
the true answers and an elastic pull toward its weights from before the loop	0.0000	0.0000

“Left” is the change at the end of practice divided by the change on return, averaged over five starts; 1 means nothing healed, and numbers above 1 mean practice pushed the learner further away.

Four lessons are in the table. **Only a record kept from before the journey brings the inside home.** A record of the hidden layer as it was before the loop leaves 0.0003 of the mark, about a thousand times less than practice alone, and an elastic pull toward the old weights leaves none that four decimal places can show. **A checker that cannot say no is worse than none.** Labelled with its own current answers, the answer term pushes nothing, and the slow pull of weight decay alone walks the learner away: its answers end 10.8 times and its inside 4.7 times further off than on return. **An honest copy of the drifted self keeps the drift.** A record taken on return, of its answers or of its hidden layer, holds the learner where the loop left it. This is the keystone of our earlier *The Anchor and the Hammer*, that a loop cannot audit its own corrupted setpoint from inside, carried into a learner: the copy is honest, and it is a copy of the wrong thing. **A stranger who knows the answers but not you rewrites you.** Practising an independent network’s answers leaves the learner’s answers nearly where the loop left them and pushes its inside 5.6 times further from home (1.1854 to 14.3292 by start). The spread is how wrong each stranger is about home: the stranger is another start’s network, and per unit of its answers’ distance from the truth the push is 148 to 226 in every start, growing in proportion to that distance (log-log slope 1.023 over the five starts; a reading after the data, runs b2 and b7, which use the same strangers).

Home is the exact answers it was raised on. How far other people’s answers push the inside depends on how far they sit from the answers the learner was raised on, not on who supplies them (run b3). A twin, raised on the same task and the same inputs from a different random start, gives answers only 0.006931 to 0.012433 away from the truth, and practising them still moves the hidden layer 1.3053 to 2.2596 times as far as the loop did. Random noise of a stranger’s size pushes 0.56 to 0.82 times as hard as the stranger’s own answers. Per unit of label change the push is 2.071 to 5.471 across twins, strangers and noise in every start. A smooth change along a direction the learner has never used is the exception, 1.2 to 12.4 per unit (run b7): the push is set by how much of the change lands on the learner’s soft directions, as the linear forecast of Act 4 predicts.

A record anyone can keep. A record of the hidden layer needs access to the inside. A record of the answers does not, and it works if it is taken on strange questions. Recording home’s answers on inputs two and a half times wider than any it trained on, and practising them alongside the true answers, cuts the inner mark on familiar ground to 0.0731 to 0.1259 of its size on return, about a third of plain practice’s, and on a different strange ground where nothing was recorded to 0.0779 to 0.1019 (run b9; Figure 6b). The record crosses mazes. Two cautions come with it. Records on wide inputs make the practice stiffer, so the stiffest network needs a smaller step: a record that blew up at the usual step stays bounded at half of it (0.2592 against 9.2083; run b16). And in one start of five, no record on wide inputs reaches the mark: a stronger one leaves 0.3454 of plain practice, and two records together destabilise it even at half the step (runs b17, b18).

A drive is held in proportion to how much of it is memory. Act 5’s records were all used after the journey. A learner can also carry something with it the whole way: a drive, a purpose it pursues alongside whatever the world asks. We gave the learner four kinds of drive, each always on and weighted like the world’s task, and measured how much of the undriven learner’s inner mark each kept after practice (runs b14, b15, b16; Figure 6c): a purpose on ground the journey never visits kept 0.88; the same purpose on the journey’s own ground 0.61; a standing wish to give again its own remembered answers 0.33; a standing record of its own hidden layer 0.010. The closer a drive is to a memory of who the learner was, the better it holds the self. A purpose that says nothing about who the learner was holds it only where it shares the road.

Holding on during the journey has a price, and it buys less. A learner that holds its inner self while the world moves follows the moving task 1.382 to 2.156 times worse than one that does not, and holding its remembered answers costs more, 1.784 to 2.821 times (run b16). This is the old trade-off between stability and adaptability in learning, here with a number on it. The same records used only after the journey cost nothing on the road and hold the self better: the remembered answers leave 0.29 of the undriven mark against 0.33 when kept on throughout (on average; better in three starts of five), and the record of the hidden layer leaves 0.00019 to 0.00085 against 0.0026 to 0.0152, better in every start by 3.1 to 65.0 times (runs b16, b17). Holding on during the trip is worse on both counts: it makes the journey harder and the homecoming less complete. Holding on for only the second half of the trip costs less than half the price (1.104 to 1.480 times the world loss, 0.27 to 0.42 of the extra cost of holding on throughout) and brings the inside home to between the two in four starts of five (run b18; Figure 6d).

What Act 5 shows. What brings the inside home is a record of who the learner was before the journey, and the deeper the record the better: answers on familiar ground help a little, answers on strange ground more, the hidden layer itself almost completely, the weights entirely. Copies of the drifted self, checkers that cannot say no, and other people's answers do not help and often harm. The engineering rule is short: *keep a record of who you are before you leave, and use it when you come home, not while you are out.* The knobs: the record's depth (answers, hidden layer, weights), the ground it is taken on (strange inputs cross mazes), its strength (a tenth still leaves only 0.0041), when it is used (after the trip costs nothing and works best), and the step size of practice (records on wide inputs need a smaller step for the stiffest networks). The anchors themselves are known tools, among them the elastic pull of elastic weight consolidation, distillation from a teacher's answers and from hidden layers, and self-distillation; what is ours is their measurement as anchors for identity after a closed curriculum, the stranger who rewrites the inside, the drive ladder and the price and timing of holding on.

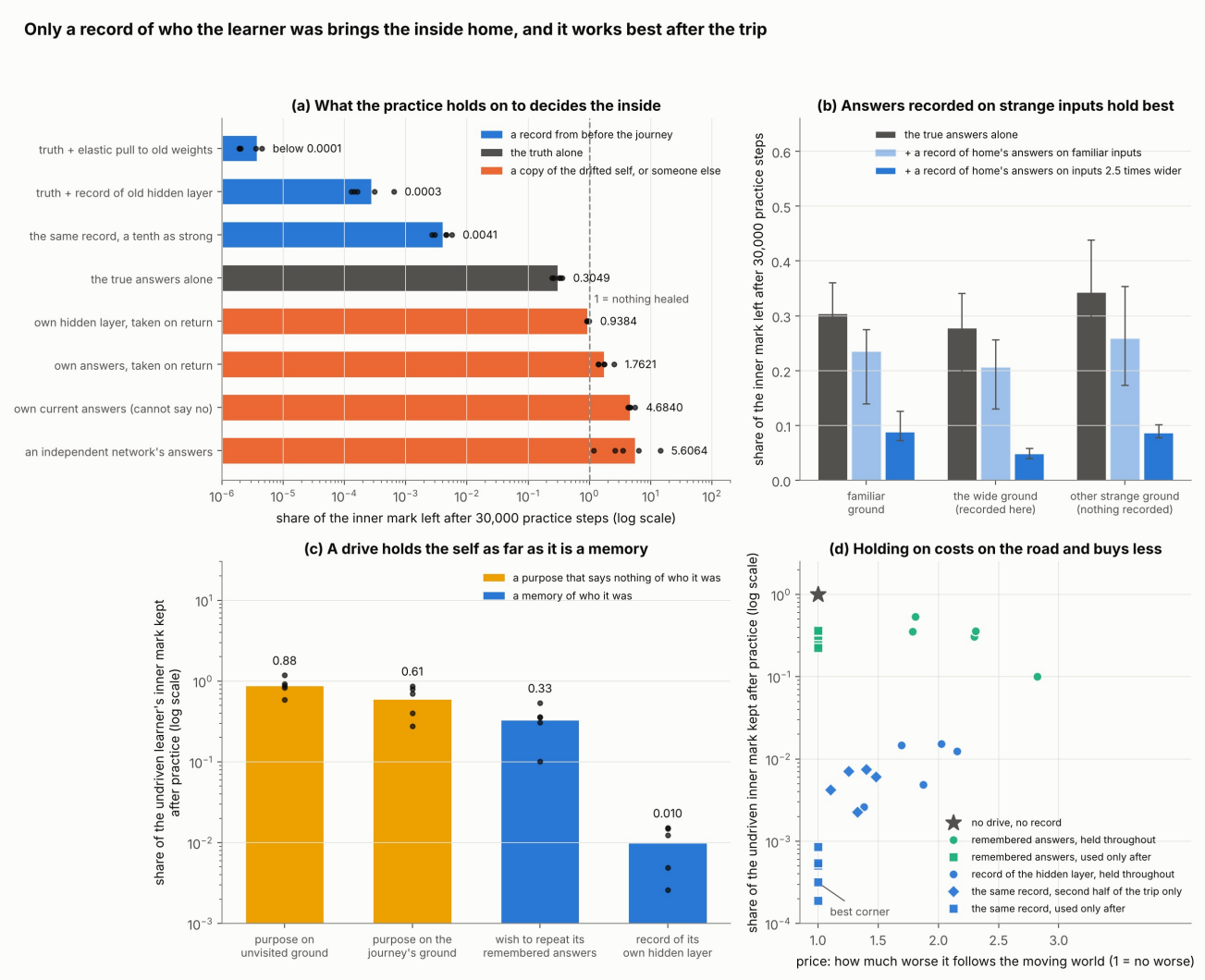


Figure 6. Only a record of who the learner was brings the inside home, and it works best after the trip. (a) The share of the inner mark left after 30,000 practice steps for the eight things practice can hold on to (bars: the mean over starts; dots: each start). An elastic pull to the old weights leaves about three millionths of it, a record of the old hidden layer 0.0003, the true answers alone 0.3049; every copy of the drifted self, and another network's answers, leave the learner where the loop put it or further away (run b2). (b) A record of home's answers taken on inputs 2.5 times wider leaves 0.0731 to 0.1259 of the mark on familiar ground and 0.0779 to 0.1019 on a strange ground where nothing was recorded; the same record taken on familiar inputs helps little (run b9). (c) Four drives carried the whole way, measured against the undriven learner: a purpose keeps 0.88 or 0.61 of the mark, the wish to repeat its remembered answers 0.33, and a standing record of its own hidden layer 0.010 (runs b15, b16). (d) What holding on costs on the road, how much worse the learner follows the moving world, against what it buys, the mark kept; each point is one start. The record of the hidden layer used only after the trip costs nothing and keeps 0.00019 to 0.00085; held throughout, it costs 1.382 to 2.156 and keeps 0.0026 to 0.0152; held for the second half only, it sits between the two on price in every start and on the mark in four (runs b16, b17, b18). Script figs/make_fig6_records.py.

Act 6: Generations, and the matriarch's note

The question. In a pod of killer whales, an old female often holds the pod's map: she leads the group through its salmon grounds, most of all in the hard years when salmon are scarce (Brent and colleagues 2015). A keeper of the map is also a single point of failure. Suppose each generation of learners walks the same journey, is born as its parent's final weights, and practises the home task before handing on. Does identity drift pile up down the generations? And can a lineage stay itself if each generation passes down only a short note about the road: one thing to seek, one thing to avoid, and the home task itself as the intent?

The lineage and the note. Every generation walks the circle loop of radius 0.2 and then practises the home task for 3,000 steps; its child starts from where it ends. Drift is measured from the matriarch, the first generation's starting point, in the hidden layer. A note is one direction in weight space and one target value along it, 97 numbers and one

more. During practice it acts as a gentle elastic pull (strength 0.05) along that single direction toward the target: that is the seek. The avoid is a brace, during the next walk of the trip, against moving along the direction the trip pushed the parent. The intent is the home task, which every generation practises anyway.

Without a note, the tail of history grows. By the sixth generation the unguided lineage has drifted 3.1994 to 4.5697 times as far as the first generation did (run b10), and by the twelfth the seed mean has reached 0.08891 (run b11), while every generation's answers stay healed. Each generation's practice heals its own trip's answers and keeps its own soft mark, and the soft marks add up.

One line from the matriarch's map holds most of it. Handing every generation the matriarch's whole map, an elastic pull toward all of her weights, holds the lineage at 0.0001 to 0.0002 of the unguided drift. A single line read from her map does nearly as much: a note whose direction is where the parent ended minus where the matriarch stood, and whose target is the matriarch's own value along it, holds the sixth generation at 0.0314 to 0.0598 of the unguided drift (run b10; Figure 7a). Ninety-seven numbers and one target remove 94 to 97 percent of the tail of history.

But a note written after rest gives the wrong advice every other generation. Under that note the drift alternates: about 0.015 in the odd generations and about 0.002 to 0.003 in the even ones, and over twelve generations the largest swings to 5.03 to 12.97 times the smallest (runs b10, b11). The reason is timing. A parent that has practised has healed; its leftover points in a direction that has nothing to do with the trip, so its child gets advice about a road it will not walk and drifts as if it had no note. That child, pulled hard by the real road, then writes a good note for the grandchild. Healed parents give wrong advice, and wounded ones give the right one.

Write it at the door. The cure is the soldier's after-action review: write the note the moment you come in from the field, before rest heals the evidence. A traveller who writes its note right after the trip, still measured from the matriarch's map, and uses it in its own practice holds the lineage level for twelve generations at about 0.0082, with the largest generation only 1.02 to 1.06 times the smallest (run b11; Figure 7b). Add the avoid brace and the lineage settles at 0.00595 by the second generation and stays there, 6.7 percent of the unguided drift at the twelfth and falling as that drift keeps growing. The avoid brace also lowers the worst generation on five new starts (to 0.613 to 0.717 of the seek note alone).

Measure it from the map, not from yourself. The same door note measured from the traveller's own birth instead of from the matriarch leaks steadily, slowing a little each generation (the twelfth adds 0.452 of what the second added), from 0.00733 in the first generation to 0.05166 in the twelfth, which is 0.581 of the unguided drift and 4.97 to 7.83 times above the map-referenced note (run b11). Measured from oneself, a lineage slowly gains on having no note at all: its drift is 0.489 of the unguided lineage's in the first generation and rises every generation, to 0.581 by the twelfth (seed means; higher at the twelfth than at the first in nine starts of ten; a reading after the data, run b11). Each generation undoes its own trip back to its own birth, and nothing ever corrects what it inherited. Reaching one generation further back, to the parent's birth, buys almost nothing: 2.5 to 5.8 percent less leak (run b13). Only the matriarch's reference stops it. This is our first paper's rule, that a mind cannot grade its own homework, showing up across generations as a slope. The map's note works only whole. Cross the two notes and both break: the chain's direction with the matriarch's target leaks to 2.942 to 4.268 times the map note's level, and the map's direction with one's own birth as target to 1.763 to 2.597 times (run b18; Figure 7d). Neither which way the note points nor what it pulls toward is enough alone; why the pair holds when neither half does is not settled.

Hand it down, and amend it at bedtime. A parent can write its door note for its child instead of using it. Because the trip pushes the same way every generation (consecutive notes point within cosines of 0.9770 of each other), a note one generation old lands the lineage on the same level as a traveller's own: 0.982 to 1.010 times by the twelfth generation (run b13). The cost of the one-generation delay is a damped settle, exactly as in a feedback loop with one step of delay: the lineage dips lowest at generation 4 or 5, overshoots a little, and settles within one percent by generation 10. A founding note for the first child halves the first swing (1.49 against 2.05 times the own-note level) but does not stop the ringing (run b14), because every child is still steering by a note one step late. What stops it is feed-forward. At bedtime the parent adds to its door note its own net change over the generation, which it already knows; that gives the child its own door note in advance, and the lineage runs within 2.84 percent of the own-note level at every generation (run b15; Figure 7c). The amendment is a dial, not a switch: half of it leaves 0.4415 to 0.4768 of the error (run b16).

What Act 6 shows. Without memory, identity drift accumulates down a lineage even while every generation's answers stay healed. A single line of memory, written at the right moment and measured from the right place, holds the lineage: write it at the door, measure it from the matriarch's map, add a brace against the road's pull, and if it is handed down, amend it at bedtime with what you know about your own change. The knobs: the note's size (one line keeps 94 to 97 percent of the full map's benefit), its timing (at the door, not after rest), its reference (the matriarch, not yourself), the avoid brace (lowers the worst generation), and the hand-down delay with its feed-forward amendment (a proportional dial). Delayed feedback that rings and feed-forward that silences it are known control engineering; what is ours is finding both inside a lineage of learners, and the size of what one remembered line can do.

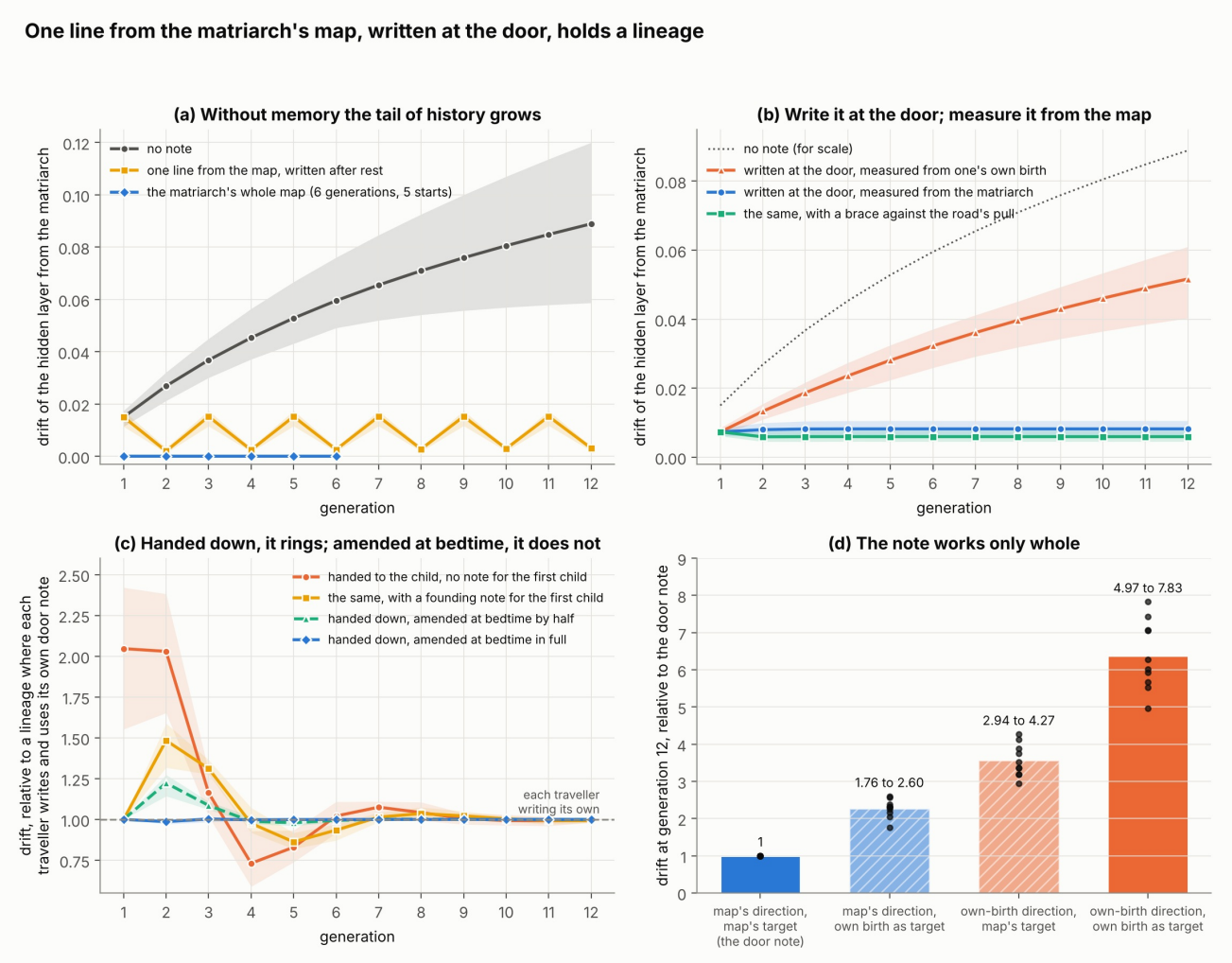


Figure 7. One line from the matriarch's map, written at the door, holds a lineage. Each panel follows the drift of the hidden layer from the matriarch, generation by generation; lines are means over ten starts and bands run from the smallest to the largest. (a) With no note the drift grows to 0.08891 by the twelfth generation. The matriarch's whole map holds it at 0.00001 (run b10, five starts). A single line from her map written after rest gives good advice only every other generation, so the lineage zig-zags between about 0.015 and 0.002 to 0.003 (runs b10, b11). (b) Written at the door, the same line holds the lineage level at about 0.0082, and a brace against the road's pull lowers it to 0.00595; written at the door but measured from one's own birth, it leaks to 0.05166, a little more slowly each generation (run b11). (c) The note handed to the child instead, relative to a lineage where every traveller writes and uses its own. With no note for the first child it rings like a feedback loop with one step of delay and settles within one percent by generation 10; a founding note halves the first swing; amended at bedtime with the parent's own change, the lineage stays within 2.84 percent at every generation, and half the amendment leaves part of the ringing (runs b13 to b16). (d) The crossed notes at generation 12, relative to the door note, each dot one start: the note works only with both its direction and its target taken from the matriarch's map (run b18). Script figs/make_fig7_lineage.py.

Act 7: The bridge to Paper 1

The question. Our first paper, *Question the Substrate*, found that a learning mind cannot grade its own homework: its own checks share its errors, and only an outside listener can hear them. Identity drift is a different kind of fault. The healed learner's answers are right; it is the learner that has changed. Can anyone outside hear that?

Graded against the truth, nobody can. Take the learner that walked the circle loop and practised for 30,000 steps, and compare it with the control that never left (run b8). On familiar inputs its answers differ from home's by 1 to 2 percent of home's own error against the truth (0.0110 to 0.0237). The strongest listener from the first paper, an outside network whose ear heard a mind's errors at +0.829 in that paper, feels the healed learner's answers within 0.14 to 1.06 percent of home's. On unfamiliar inputs two and a half times wider than any it trained on, the learner's answers move 9.07 to 14.46 times further from home's than on familiar ground, but home's own error there grows 23.96 to 59.29 times, so against the truth the change is quieter on new ground, not louder. The first paper's retired

“comma” instrument, the gap between two small lessons taken in opposite orders, reads the inside at the inside’s own size (0.643 to 1.119 times the hidden change) and no louder. Errors need an outside ear. Identity drift makes no error for an ear to hear.

Graded against a kept copy of home, the mark is loud as soon as the checker leaves home. An independent run, designed, registered and run in the first paper’s own session (six predictions, six held), asked the same question with a different yardstick: compare the healed learner with a kept copy of home, the control, and measure the gap against the scale of home’s answers (run bridge_b1; Figure 8a). At home the gap is 0.043 percent. On inputs three times wider it is 8.888 to 9.876 times larger, and six times wider 9.425 to 15.57 times. Give both networks the same new lesson to learn and their answers at home draw apart, 2.801 to 9.356 times the gap before the lesson. A random change of the same home gap, spread over all directions, stays quiet away from home (1.01 to 2.085 times) and fades under a lesson (0.2789 to 0.6707). The two runs were written separately in two sessions; on the five healed networks they share, they print the same hidden change and the same home gap to every digit. Read together, they say that the yardstick decides. A check graded against the truth cannot hear identity drift. A check graded against a kept copy of home hears it once it leaves home, which is the detector form of the record that Act 5 used as an anchor.

Why a new lesson wakes the mark. The lesson checker turns out to be a linear instrument: a first-order model that carries the mark along the control’s own lesson and reads it with the control’s own sensitivity matches the measured amplification within 0.38 to 2.5 percent at every lesson length (run b12; Figure 8b). Split that model in two and the reason appears. Read the carried mark with home’s old sensitivity, or the old mark with the carried sensitivity, and each half alone is 2.18 to 5.68 times louder than the whole (Figure 8c). The lesson moves the blind spot and the mark together, mostly in step, and the checker hears only where they fall out of step. By 3,000 lesson steps they are back in step and the mark is hidden again (1.156 to 1.754 times its home gap). Because the lesson uses the same inputs as home, everything it adds to the curvature comes from the change of labels times the network’s second derivative, and along the loop’s mark that added curvature is 77 to 235 times the home curvature (run b12).

It hears softness, and big lessons hear best. The isotropic random change in the independent run was mostly stiff, which is why it had to be so small to be quiet at home. A random change drawn only from the soft directions, scaled to the same home gap, is about as loud off home as the loop’s mark (0.5965 to 1.005 of it at three times the input width) and at least half as loud under a lesson (0.654 to 0.974 with 25 such changes per start; runs b12, b13). So the checker hears any drift that the answers have hidden; it cannot by itself say where the drift came from. And bigger lessons hear louder: a lesson whose target turns by a right angle or more beats a 45-degree lesson in 14 of 15 starts (runs b14, b16, b17; Figure 8d), reaching 4.465 to 172.8 times the home gap on five of them. When the loudest moment comes is not simple: four registered attempts to tie it to the control’s settling time missed, and that reading is laid down.

What Act 7 shows. The two papers meet on one rule. Errors need an outside ear, because a mind’s own checks share its errors. Identity needs a kept record, because a changed mind gives the right answers: graded against the truth it is silent, and graded against a copy of who it was it is loud, as soon as the check asks strange questions or sets a new lesson. The same kept record that brings the self home (Act 5) is the one that can tell it has drifted. The knobs: the yardstick (truth or a kept copy of home), the distance from home of the questions asked (the mark grows about tenfold three input-widths out), and the size of a new lesson (big lessons hear best). The cautions: the checker hears any soft difference, so it detects drift without naming its cause, and the lesson’s loudest moment has no simple clock.

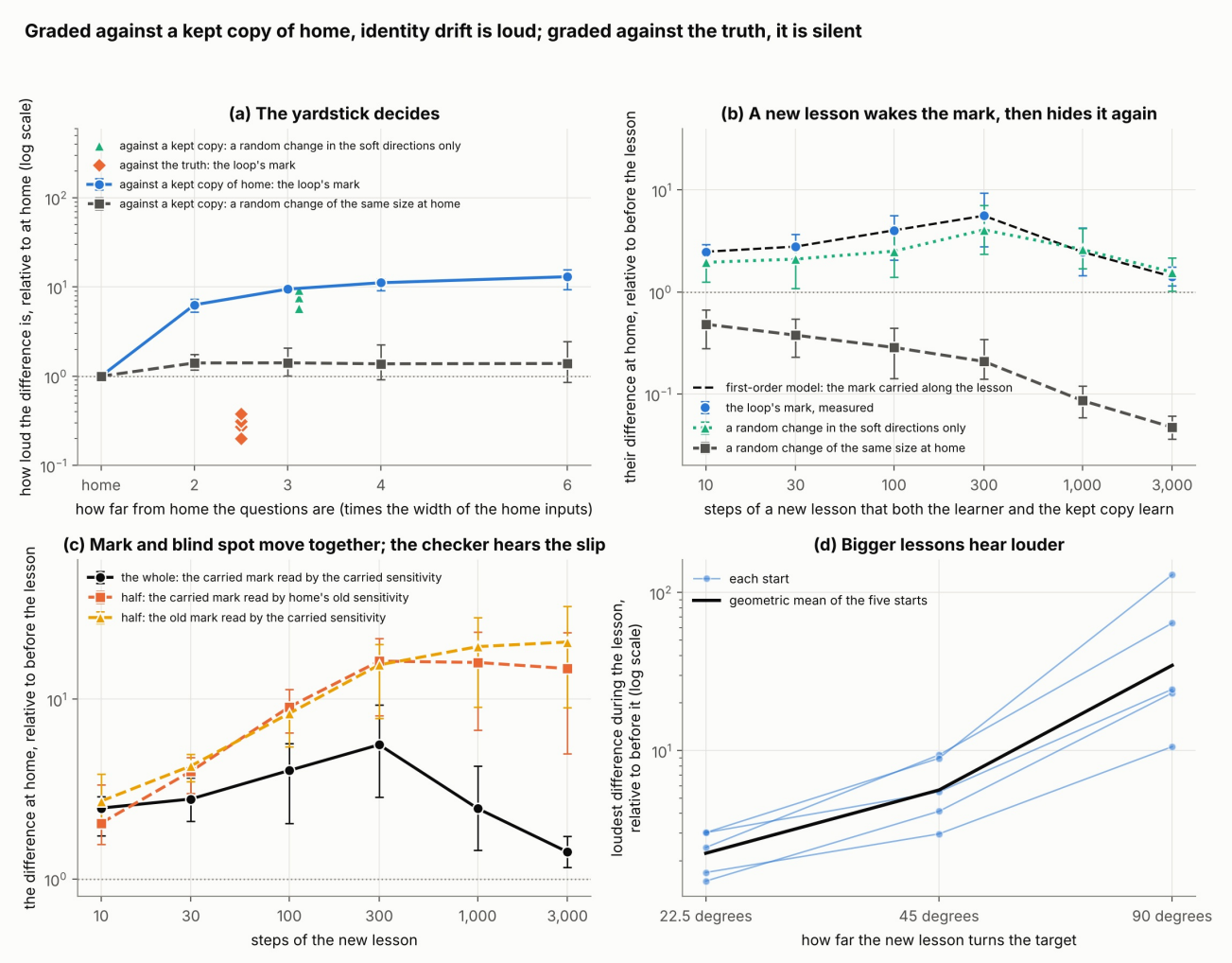


Figure 8. Graded against a kept copy of home, identity drift is loud; graded against the truth, it is silent. (a) How loud the healed learner's difference is away from home, relative to at home. Against a kept copy of home, the loop's mark grows 8.888 to 9.876 times by three input-widths and 9.425 to 15.57 times by six, while a random change of the same size at home stays near its home level (0.855 to 2.466); a random change drawn only from the soft directions is about as loud as the loop's mark (5.826 to 9.295 at three widths). Against the truth, on inputs 2.5 times wider, the loop's mark is quieter than at home (0.20 to 0.38), because home's own error grows faster than the mark (runs bridge_b1, b8, b12). (b) The difference at home while the learner and the kept copy learn the same new lesson: the loop's mark wakes to 2.801 to 9.356 times its home level by 300 steps and is nearly hidden again by 3,000; a first-order model that carries the mark along the lesson follows it within 0.38 to 2.5 percent at every length; a random change of the same home size fades (runs bridge_b1, b12). (c) The model's two halves, the carried mark read with home's old sensitivity and the old mark read with the carried sensitivity: each peaks 2.18 to 5.68 times higher than the whole, because the lesson moves the mark and the network's blind spot together and the checker hears only where they slip apart (run b12). (d) The loudest difference during the lesson against how far the lesson turns the target: 1.493 to 3.026 times at 22.5 degrees, 2.970 to 9.386 at 45, 10.58 to 129.6 at 90 (run b12; runs b14, b16 and b17 extend the comparison to 15 starts). Bars run from the smallest to the largest of five starts (script figs/make_fig8_bridge.py).

Act 8: The engineering knobs

The question. Everything in Acts 1 to 7 can be read as a dial. If you are training a learner through a changing world, or passing one down a line of learners, and you care that it stays itself, which dials do you have, which way do they turn, and in what order should you set them?

The knobs, with their numbers.

Knob	Which way it turns	The number behind it	Act
Loop area	The direction-flipping mark grows in proportion to the area enclosed	about 0.12 weight units per unit of area	1
Loop size and shape	The direction-blind mark grows with the square of the size; out-and-back marks most	0.16007 for out-and-back against 0.014719 for the circle’s area part	2
Loop speed	A best speed for leaving a mark; faster cuts corners, slower lets the learner follow	peak near 2,000 to 4,000 steps per lap	1
Pace asymmetry	An uneven pace leaves an imprint that never heals	rho 0.0661 to 0.3348 after 30,000 steps	2
Step size	Does not matter at fixed learning time	ratios 0.9966 to 1.0003	2
Playback order	Walking back erases the area part and doubles the blind part	cosine 0.9976 to 0.9993 with the blind part	2
Loop topology	A half turn of a direction-blind task trades two units for good	68 to 85 percent of the distance is a relabelling	3
Practice length	Answers heal fast, the inside slowly	0.0167 against 0.3049 left after 30,000 steps	4
Record depth	Deeper records hold better: true answers alone < home’s answers on strange ground < the hidden layer < the weights	inside left 0.3049; 0.0731 to 0.1259; 0.0003; 0.0000	5
Record timing	Use the record after the journey, not during it	0.00019 to 0.00085 after, against 0.0026 to 0.0152 during	5
Holding during the journey	Costs adaptability	1.382 to 2.156 times the world loss	5
Note timing (lineage)	Write the note at the door, not after rest	level within 1.02 to 1.06, against swings of 5.03 to 12.97	6
Note reference (lineage)	Measure from the matriarch, not from yourself	chain leaks to 4.97 to 7.83 times higher by generation 12	6
Hand-down amendment	Add your own net change at bedtime; a proportional dial	within 2.84 percent; half leaves 0.4415 to 0.4768 of the error	6
Check yardstick	Grade against a kept copy of home, not against the truth	silent against the truth; 8.888 to 9.876 times louder three input-widths out	7
Check lesson size	Bigger new lessons hear the mark louder	the louder big lesson beats a 45-degree lesson in 14 of 15 starts	7

The manual, in the order you would use it.

1. **Before the journey, take a record.** Record who the learner is at home: its hidden layer if you can reach the inside, its weights if you can store them, or at the least its answers to strange questions, which need no access to the inside and still cross to ground the record never saw (Act 5).
2. **Shape the journey if you can.** A closed loop of tasks always leaves a mark. Smaller loops leave smaller marks, in proportion to their area for the part that knows the direction and to the square of their size for the rest. Going out and coming straight back marks the learner more than going round. Keep the pace even, because an uneven pace leaves an imprint that practice never removes (Acts 1 and 2).
3. **Watch for half turns.** If the task does not care which way a direction points, half a turn hands the learner back its own task with the frame flipped, and two of its units will trade roles for good once the turn passes a quarter of the way. No check on answers can see this; only a record that names the parts can (Act 3).
4. **During the journey, let the learner adapt.** Holding the self in place while the world moves makes the learner follow the world 1.4 to 2.2 times worse and brings it home less completely afterwards (Act 5).
5. **On return, forecast before you practise.** Practice is linear relaxation, so the size and place of what will remain can be computed at the moment of return from the leftover and the network’s curvature. Expect the answers to heal within a few thousand steps and the inside to keep about a third of its mark after 30,000 (Act 4).
6. **Then use the record.** Practise the home task with a gentle pull toward the record taken before the journey. A record of the hidden layer used this way leaves a few ten-thousandths of the mark. Do not practise against a copy of the learner taken on return, against its own current answers, or against another network’s answers: each keeps the drift or makes it worse (Act 5).

7. **If the records are on wide inputs, mind the step.** Records on inputs far from home stiffen the practice; for the stiffest networks halve the step size and double the steps (Act 5).
8. **To check whether the learner is still itself, do not ask it questions you know the answers to.** Compare it with the kept copy of home, on questions far from home or after a big new lesson that both must learn. Expect to detect any hidden drift that way, but not to learn where it came from (Act 7). A check on answers alone catches some failures and is blind to others: a checker that cannot say no walks the answers away faster than the inside (10.8 times against 4.7), so the answers show it, but practising a stranger's answers leaves the answers nearly where they were (0.97) while pushing the inside 5.6 times further from home, and only a record of the inside sees that (Act 5).
9. **For a line of learners, pass down one line.** Have each generation write a one-line note the moment it returns, measured from the first generation's map, brace against the road's pull on the next walk, and when the note is handed down, amend it at bedtime with the parent's own net change (Act 6).

What Act 8 shows. The results of this paper are a working kit. Identity drift has measurable causes (area, shape, pace, topology), a predictable course (linear relaxation), a cure (a record kept before the journey, used after it), a test (a kept copy of home asked strange questions), and a way to last across generations (one well-timed line from the matriarch's map). None of the knobs needs the learner to understand itself; each needs someone to have kept a record of who it was.

Limits

The learner is small and the world is synthetic: a network of 97 weights (4 inputs, 16 tanh units, 1 output), one family of regression tasks, full-batch gradient descent with a small weight decay, and no sampling noise. Most results rest on five random starts; the lineage results on ten, and the lesson checker on twenty. Every number is a range over those starts, not a confidence interval. One start of five, start 3, is the stiffest network (largest curvature 7.1646, and the fewest soft directions, 65 against 73 to 79; runs b16, r1), and it is the outlier in several places: records on wide inputs destabilise it at the full step and do not reach its mark at half the step (runs b16, b17, b18), holding a record for half the journey leaves it further off than holding on throughout (run b18), its marks scatter at the largest loop (run s1), and its clones split most under symmetry-breaking noise (run b1 T5). The ranges over five starts include it; where it is the only exception, the text says so. Readings computed after the data are labelled as such. The independent bridge run was designed and run in the first paper's own session, so it is an independent code path within one project, not an outside replication. This is a preprint that has not been peer reviewed, and its search of the literature is limited: each cited work was checked against its publisher's record and abstract, and most were not read in full. Every number in the paper is printed by a tool that ships with it.

Open questions

Each says why it is here and whether we can run it.

Question	Where it stands	Can it be run here?
What makes the rest of the direction-blind mark (Act 2)?	Linear response to the path explains part of it (cosines 0.50 to 0.72); the rest should be the learner's changing curvature along the way	Runnable (the re-linearised forecast applied to the journey); not run, since the paper's experiments are closed
Why does the matriarch's note hold when neither its direction nor its target does alone (Act 6)?	Crossing them breaks both (2.942 to 4.268 and 1.763 to 2.597 times the map note)	Open
Can anything reach seed 3's mark (Act 5)?	Records on wide inputs either fail to reach it or destabilise it, even two together at half the step	Runnable at a quarter step; not run
Can the traded units be named before a half turn in every start (Act 3)?	With clones merged, in four starts of five	Runnable with a looser clone tolerance; not run
When is a new lesson loudest (Act 7)?	Four registered attempts to tie it to the settling time missed	Open; no simple law found
Do the results hold in larger learners, with stochastic training, on real tasks?	Everything here is one small learner	Not here; it needs larger runs and is left for others
What if the learner chooses its own drive, for example which lesson to practise?	Every drive here was given	Not in this paper; it belongs to its own

Answered before publication (run b18): the forecast of Act 4, re-linearised along the way, holds for large and unusual label changes (9 of 10 seed-cases within 25 percent); holding a record for the second half of the journey costs less than half the extra price of holding it throughout (journey loss 1.104 to 1.480 against 1.382 to 2.156) and brings the

inside home to between “during” and “after” in four starts of five.

Related work, and what is ours

Known, and reproduced or confirmed here. The geometric phase of parallel transport (Berry 1984; Hannay 1985; Gauss-Bonnet; the Foucault pendulum), which the hand-built agent reproduces. Complementary filtering, lead compensation, loop delay and feed-forward from control engineering, and the inertial-navigation logic of anchors. The order effect between two lessons, the step size squared times their Lie bracket (Rukhovich, Podolskiy and Piontkovskaya 2025; Dherin and colleagues 2025; the related implicit-regularization analysis of Smith, Dherin, Barrett and De 2021), of which the area law is the closed-loop form. The stability-plasticity trade-off in learning.

Known in kind, new in place or number. Holonomy of a fixed network’s representation over loops of inputs (Sevetlidis and Pavlidis 2026), which names curricula as a loop family for the future; here a learning network walks one. Representational drift with stable behaviour (Driscoll and colleagues 2017; Rule, O’Leary and Harvey 2019). Sloppy models (Gutenkunst and colleagues 2007) and neuron condensation (Luo, Xu, Ma and Zhang 2021). Anchors used in continual learning and distillation: elastic weight consolidation (Kirkpatrick and colleagues 2017), L2-SP (Li, Grandvalet and Davoine 2018), distillation from answers and from hidden layers (Hinton, Vinyals and Dean 2015; Romero and colleagues 2015), self-distillation (Mobahi, Farajtabar and Bartlett 2020) and learning without forgetting (Li and Hoiem 2018). Permutation symmetry and the realignment of hidden units (Ainsworth, Hayase and Srinivasa 2023). Underspecification, where models that agree in-distribution diverge under stress (D’Amour and colleagues 2022), and relearning tests in machine unlearning, where a little fine-tuning brings back knowledge that an unlearning method had only hidden (Deeb and Roger 2025). Swimming at low Reynolds number, where a cyclic change of shape gives a net displacement set by the area of the loop (Shapere and Wilczek 1989), an analogue of the area law in overdamped motion.

Ours, as far as the reads go. The area law measured in a learner’s own weights and hidden layer, with the turned-copy and time-reversal design that separates it from lag; the larger direction-blind mark, its pace imprint and its doubling under playback; the discrete identity drift of the half turn, one event at a quarter turn, and its mirror pairs; answers healing 18.49 times better than the inside, with the inside’s fate forecast at return; the anchor table, including the stranger who rewrites the inside and the copy that keeps the drift; the drive ladder, the price of holding on during a journey and the rule to use the record after it; the lineage results, from the matriarch’s single line to the bedtime amendment; and the bridge to the first paper, where the yardstick decides whether identity drift can be heard.

The author’s own lineage. *The Anchor and the Hammer* (concept DOI 10.5281/zenodo.21450883) argued that a reference-tracking loop cannot audit its own corrupted setpoint from inside; Act 5 carries that keystone into a learner. *Question the Substrate: Why a Learning Mind Can’t Grade Its Own Homework* (concept DOI 10.5281/zenodo.24094163) is the first paper of this pair; Act 7 is the bridge between them. *The Loop Atlas* (concept DOI 10.5281/zenodo.24100672) calls the number that turns the room a loop encloses into leftover the loop’s *grip*; this learner’s grip is about 0.12 weight units per unit of enclosed task area (Act 1), and Act 5 adds that a drive grips the self only as far as it is memory.

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Appendix A: every registered prediction and its verdict

Every prediction below was written into a PREDICTIONS file before the run that tested it (the files are named in the Declaration's provenance table), and every verdict is printed by the run itself. In all, 164 predictions were registered and tested on their first run: 81 held and 83 missed. In the first two studies, 11 redesigned predictions were tested again after a miss, and 4 of them held; from batch 1 on, every redesign was registered as a new prediction in a later batch and is counted among the 164. The Paper 1 session's independent bridge run registered 6 more, and 6 held. A HELD verdict says that the registered bar was met; it does not say by how much. Table A1 lists every prediction by its ID; Table A2 says, part by part, what was written first and what came. Appendix B tells the misses that changed the story.

Table A1. Every registered prediction by ID, with its verdict.

Part	Tested	Held	Held (IDs)	Missed (IDs)

The hand-built agent (runs q1 to q5)	25	20	P1.1, P1.2, P1.3, P1.4, P1.5, P2.1, P2.2, P2.5, P2.6, P2.7, P3.1, P3.2, P3.3, P4.1, P4.2, P4.3, P4.4, P4.5, P4.6, P5.2	P2.3, P2.4, P3.4, P5.1, P5.3
The hand-built agent (runs q1 to q5): redesigns tested again	3	3	P2.3r, P2.4r, P3.4r	-
The first learner study (runs s, r and f)	12	5	P1, P2, P3, P4, P5	P6, P7, P8, P9, P10, P11, P12
The first learner study (runs s, r and f): redesigns tested again	8	1	R9	F1, R6a, R6b, R8a+R8b, R8c, R11, R12
Batch 1: what is the direction-blind part of the mark?	12	5	T1b, T4a, T4b, T5a, T6b	T1a, T2a, T2b, T3, T5b, T6a, T7
Batch 2: anchors in a learner	9	7	H1, H2, H3, H4, H5, H6, H7	R1a, R1b
Batch 3: the outsider dial	5	2	D1, D5	D2, D3, D4
Batch 4: the Mobius loop and the twin	7	3	M3, M4, M6	M1, M2, M5, T1
Batches 4 and 5: the Mobius loop and the twin in the mirror	5	1	A4	A1, A2, A3, A5
Batch 6: when does the Mobius swap happen, and does a second half turn undo it?	4	1	S1	S2, S3, S4
Batch 7: the no-practice forecast, the label-distance law on new sources, and the mirror pairs	7	1	F1b	F1a, L1, L2, L3, P1, P2
Batch 8: the bridge test	5	3	B1, B4, B5	B2, B3
Batch 9: time reversal, the kept record on strange ground, and waypoints played back	10	2	E2, K3	E1, K1, K2, K4, W1, W2, W1r, W2r
Batch 10: the matriarch’s note	6	4	G1, G2, G3, G6	G4, G5
Batch 11: the after-action review written after the action	5	3	N1, N4, N5	N2, N3
Batch 12: why a new lesson wakes a hidden mark	8	5	Z1, Z2a, Z3, Z4, Z7	Z2b, Z5, Z6
Batch 13: the parent writes the review at the door	8	4	J2, J5, J6, Z5r	J1, J3, J4, Z6r
Batch 14: a learner with a drive of its own	9	3	V4, V5, J4r	V1, V2, V3, J1r, J1r2, Z6r2
Batch 15: a drive that is a memory	8	4	W4, W5, J1rr, J1rr2	W1, W2, W3, Z6r3
Batch 16: the bottom of the drive ladder, the price of holding on, and three earlier redesigns	9	5	X1, X2, X4, J1h, K4r	X3, X5, R1, K1r
Batch 17: batch 16’s three redesigns	3	2	Y1, R1r	K1rr
Batch 18: the six runnable	7	1	Q4	Q1, Q5, Q2a, Q2b, Q3, Q6

open questions				
The Paper 1 session’s independent bridge run	6	6	BP1, BP2, BP3, BP4, BP5, BP6	-

Table A2. What was written first and what came. The first part lists the predictions that held, the second those that missed or were void, with the redesign that later landed where there was one; part names follow Table A1.

IDs	Part	What was written first	What came	Run
P1.1 to P1.5	prototype	agent does the task; blend wins only with noise; speed gap at least 15x	all held; speed ratio 27.5	q1
P2.1, P2.2, P2.5, P2.6, P2.7	prototype	flat drift 0; follower heals; triangle 90; laps add; speed	all held	q2
P3.1 to P3.3	prototype	no anchor 120 to 175; real under 15; fakes identical to none	147.070; 3.106; 0.00e+00	q3
P4.1 to P4.6	prototype	interior peak; moves with noise; not 90; interior best inner speed	all held	q4
P5.2	prototype	outer dominates for 1 to 16 rounds	held, by negative weights	q5
P2.3r, P2.4r	prototype	redesign: exact rotation rule	spread 0.000000; reverse sum 0.000000	q2r
P3.4r	prototype	redesign: anchors every 10 steps	largest report 3.9452	q3r
P1 to P5	learned	area law in weights and hidden layer; one direction; lag leaks in without the turned copy	all held	s1
R9	learned	heal run with turned-copy average	hidden 0.675 at 3,000; 0.3049 at 30,000	r3
T1b	batch 1	fast/slow mixed pace sits between the symmetric paces	5 of 5 starts	b1_t1
T4a, T4b	batch 1	rest heals by linear relaxation	gaps 0.0442 (share), 9.10% (size)	b1_t4
T5a	batch 1	clone share grows as noise squared	slopes 1.9335 to 2.0064	b1_t5
T6b	batch 1	lobe swap reverses A at least partly	cosine -0.3365 to -0.6561	b1_t6
H1 to H7	batch 2	record anchors restore the inside; answer, mirror, copy and outsider anchors do not	inside left 0.0003 with the hidden-layer record and 0.0000 with the elastic pull; 0.9384 to 5.6064 with the copies and the stranger (Act 5’s table)	b2_anchors
D1, D5	batch 3	push grows with the outsider’s distance; distance ranks the starts	every start; Spearman 1.0000	b3
M3, M4, M6	batch 4	out-and-back comes home; Mobius answers come home; Mobius never heals	raw 0.1238 to 0.5883; probe loss at the control’s; flat from 3,000 to 30,000	b4_mobius
A4	batch 5	the out-and-back has nothing to relabel	ratio 1.0000 in every start	b5
S1	batch 6	the swap happens at one place near the middle	first at the fifth eighth in every start	b6
N1, N4, N5	batch 11	after-action notes remove the rhythm; avoid lowers the worst generation on new starts; chain-referenced after-action notes drift away	1.02 to 1.06; 0.613 to 0.717; 4.97 to 7.83 times	b11
G1, G2, G3, G6	batch 10	drift accumulates without a note; the map holds; one map direction holds most; answers stay healed	3.1994 to 4.5697 times; 0.0001 to 0.0002; 0.0314 to 0.0598; at most 0.00579	b10
E2, K3	batch 9	time-reversal echo lies along the blind part; a wide record crosses to the x4 maze	cosine 0.9520 to 0.9768; 0.0779 to 0.1019 vs 0.2920 to 0.4384	b9
B1, B4, B5	batch 8	answer checker deaf; comma reads the inside at its own size; Paper 1’s outside ear deaf	0.0110 to 0.0237; 0.643 to 1.119; 0.0014 to 0.0106	b8

F1b	batch 7	the forecast holds for the largest label changes in 3 of 5 starts	0.943, 1.025, 0.774	b7_forecast
Q4	batch 18	re-linearised forecast within 25% in 4 of 5 for SMOOTH and for OUT-4	5 of 5; 4 of 5	b18
Y1, R1r	batch 17	inner record after the trip only below 0.02; the louder big lesson beats pi/4 in 4 of 5 new starts	0.00019 to 0.00085; 5 of 5	b17
X1, X2, X4, J1h, K4r	batch 16	inner drive below 0.1 after practice and 0.5 at return; area law survives it; half amendment leaves 0.3 to 0.7 of the error; start 3’s x4 record bounded at half step	0.0026 to 0.0152; 0.2002 to 0.4086; 0.9619 to 0.9860; 0.4415 to 0.4768; 0.2592	b16
W4, W5, J1rr, J1rr2	batch 15	remembered drive beats the assigned one on shared ground; area law survives both drives; the amended note is level and tracks the own-note lineage	0.3676 to 0.7710; 0.9758 to 1.0263; 1.028 to 1.094; at most 0.0284	b15
V4, V5, J4r	batch 14	the area law survives a drive; the world heals as well with one; the chain leak lies across the note	0.9654 to 1.0175; 0.8691 to 1.1722; 0.93101 to 0.99023	b14
J2, J5, J6, Z5r	batch 13	parent’s door note within 1.5x of your own at gen 12; chain still above 1.5x map; avoid helps; soft noise at least half as loud (25 marks)	0.982 to 1.010; 4.869 to 7.562; 0.682 to 0.885; 0.654 to 0.974	b13
Z1, Z2a, Z3, Z4, Z7	batch 12	first-order lesson model within 10%; each half at least 0.25 of the whole; label-change curvature at least 10x home’s along the mark; soft noise at least half as loud at distance; louder with the angle	0.003762 to 0.02503; 2.18 to 5.68; 77.02 to 234.8; 0.5965 to 1.005; 1.493 to 3.026, 2.97 to 9.386, 10.58 to 129.6	b12
BP1 to BP6	bridge, independent (Paper 1 session)	home checker blind (under 1%); distance dial at least 3x at d = 3 and rising; a lesson at least 2x; loop mark at least 3x and 2x a matched random mark’s	0.03435% to 0.0571%; 8.888 to 9.876, rising every step; 2.801 to 9.356; ratios 4.441 to 9.673 and 4.177 to 19.72	bridge_b1

IDs (missed or void)	Part	What was written first	What came	Redesign that landed
P2.3, P2.4	prototype	start spread under 0.1 deg; reverse sum under 0.1	2.553038; 2.804638	P2.3r, P2.4r
P3.4	prototype	copy reports under 5 deg	7.148	P3.4r
P5.1	prototype	zero rounds: score flat in psi	range 0.0705	-
P5.3	prototype	blend helps only from 64 rounds; psi near 41 at 256	helps at 0 (negative weights); psi 91 in a still world	(q5r tables: 64 rounds and 41 deg in a moving world; no pass/fail line)
P6, P7	learned	direction flips the sign; flat loop leaves little	cosine +0.8656 to +0.8862; ratio 8.908 to 9.767	- (R6a, R6b, F1 missed)
P8	learned	figure eight leaves little	void: A zero by construction	- (R8a+R8b, R8c missed)
P9, P10	learned	heals in function, not representation	hidden 0.1838 left (first-order lag inflated the reference)	R9
P11	learned	laps add	1.1119 in one start vs 1.1 cap	- (R11 missed)
P12	learned	speed hardly matters	ratios 2.6894 to 19.2881	- (R12 missed)
T1a	batch 1	0.05 < rho < 0.5 at return	0.5447 to 0.6189	redesign registered next (rest 10,000 and 30,000)
T2a, T2b	batch 1	one sweet-spot pace	the mean hid two start families	redesign: new starts 5 to 9

T3	batch 1	middle band ranks sensitivity (expected miss)	Spearman 0.3000	same new-start redesign
T5b	batch 1	clone share < 0.05 at eps 1e-2	start 3 0.09217	redesign: clone-difference curvatures
T6a	batch 1	lobe swap cosine < -0.8 (expected miss)	-0.3365 to -0.6561	none needed
T7	batch 1	64,000 steps lowers A in every start	start 0 ratio 1.1385	-
R1a, R1b	batch 2	the persistent mark forgets the pace (rho < 0.1)	0.0661 to 0.3348; 0.0917 to 0.3796	none needed: the permanent imprint is the result
D2	batch 3	push linear in label distance, slope 0.8 to 1.2	start 4 0.6721	redesign: k = 0.25, 0.5, 1
D3	batch 3	a twin pushes less than a stranger	twin 1.7571 vs stranger 1.1854 in start 0	redesign: label-distance law on new sources; linear-response forecast
D4	batch 3	noise pushes under half as much	0.56 to 0.82	same redesign as D3
M1, M2, M5	batch 4	Mobius returns the input mirror; full turn returns home; recognizes itself in the mirror view (raw frame)	raw frame cannot see relabelling	b5 (frame-aligned) found the swap
T1	batch 4	twins are different minds (aligned > 0.5 raw)	ratios 0.2011 to 0.6415	none: start-dependent, reported
S2, S3, S4	batch 6	second half turn swaps back; reverse swaps the same pair; leftover flips with direction	3 of 5; 3 of 5; cosines +0.1337 to +0.9871	redesign: predict exchanged pairs from each network's plus/minus unit pairs
N2, N3	batch 11	after-action notes halve the worst generation; the map note after rest stays within 25%	0.41 to 0.90; 1.14 to 2.43 times	covered: parent-to-child after-action version; after-action map note is level
G4, G5	batch 10	a chain note fails; the avoid helps at generation 6	0.0446 to 0.1407 of NONE; worse at gen 6	redesigns: 12 generations; worst generation on new starts; write the note before practice
E1, K1, K2, K4, W1, W2, W1r, W2r	batch 9	echo within 0.25 of 2S; wide record below 0.3 of plain; familiar record adds nothing; x4 record crosses; crossing waypoint matters most (no rest; with rest)	0.2504 to 0.3387; 0.35 in two starts; 0.53 to 0.92 of plain; start 3 blew up; 2 of 5; 3 of 5	redesigns: two records; half step for start 3; covered or none for the rest
B2, B3	batch 8	the comma hears louder than answers; new ground makes the mark louder	0.18 to 0.64 of answers; 0.20 to 0.38 of familiar	redesigns: comma specificity; kept record of home's answers on new ground
F1a, L1, L2, L3, P1, P2	batch 7	forecast within 25% everywhere; smooth pushes like a mind; noise 0.4 to 0.8; proportional; mirror pairs; exact mirrors in starts 1 and 3	37 of 45 within 25%; 0.290 to 3.410; up to 1.003; up to 1.2387; 3 of 5; none	redesigns: second-order forecast term; soft-share ranking; clone-merged pairing
Q1, Q2a, Q2b, Q3, Q5, Q6	batch 18	blind mark is linear response; crossed notes hold or leak by reference; two records reach start 3; clone-merged pairing closes everywhere; half-hold lands between	cosine 0.50 to 0.72; 2.942 to 4.268 and 1.763 to 2.597; start 3 3.5004; 4 of 5; 4 of 5	written down, not run (science closed): see B18_RESULTS
K1rr	batch 17	start 3's doubled wide record at half step below 0.3 of plain practice	0.3454	closed after two redesigns; the two records together were tested in batch 18 (Q3, missed)
X3, X5, R1, K1r	batch 16	after-trip drive within 10% of always-on; holding costs at least 2x; pi/2 loudest in 4 of 5 new	0.420 to 3.349; 1.382 to 2.156; 3 of 5; start 3 blew up	inner drive after the trip only; none (measurement stands); big-lessons-beat-pi/4 on starts 15 to

		starts; doubled wide record below 0.3		19; start 3 doubled record at half step
W1, W2, W3, Z6r3	batch 15	remembered drive below 0.5 after practice and 0.7 at return; assigned drive on shared ground below 0.5; peak over settling in 0.7 to 1.5 at 3pi/4 in 8 of 10	0.1009 to 0.5351; 0.6129 to 1.0694; 0.2746 to 0.8665; 3 of 10	inner record as a drive; drive on only in practice; Z6 closed after the fixes tried
V1, V2, V3, J1r, J1r2, Z6r2	batch 14	a drive holds the inner arrow at return and halves it after practice; at least 10 fewer soft directions; a founding note levels the lineage and tracks the own-note arm within 10%; peak over settling within 0.7 to 1.5 in 12 of 15 new cases	0.7609 to 1.0124; 0.5896 to 1.1796; 4 to 8; 1.402 to 1.856; 0.3174 to 0.5880; 5 of 15	drive on shared ground (second input); lead-corrected note (twice the parent's minus the grandparent's); the law at 3pi/4 on ten starts
J1, J3, J4, Z6r	batch 13	parent's door note level (under 2x) over gens 2 to 12; beats the after-rest note's worst; reaching back to the parent's birth cuts the leak below 0.75x; peak over settling agrees across angles within 25%	1.851 to 3.972 (start-up settle; 1.057 to 1.199 from gen 6); 0.819 to 1.367; 0.942 to 0.975; 1.216 to 33.63	founding note for generation 1; along-or-across split of the chain leak; the loose law on starts 5 to 9
Z2b, Z5, Z6	batch 12	each half below 0.75 of the whole; soft noise at least half as loud under a lesson; peak step within 2x across angles	each half 2.18 to 5.68 times the whole (they cancel); start 3 0.4863; pi/2 2.444 to 7.973 times later	none for Z2b (the cancellation is the result); 25 soft marks per start; peak as a fraction of each lesson's settling time
A1, A2, A3, A5	batch 5	aligned < 0.3 raw (Mobius, full turn, mirror*); realigned hidden < 0.3	0.1499 to 0.3219; up to 0.7997; up to 0.3812; up to 0.33	redesign: track the swap along the loop; half turn twice

Appendix B: the misses that changed the story

Most of what this paper says was found by a prediction that missed. Each row below is a miss that changed the story: what was written down before the run, what the run printed, and what the paper became because of it. The full list, with every verdict, is Appendix A.

Table B1. The misses that changed the story, in the order they happened.

Miss	Written first	What came	What it changed
P9, P10 (first learner study)	Practice heals the answers and not the inside, measured on single runs	Only 0.1838 of the inside seemed left, because the learner's lag behind the moving task inflated the reference	Each loop is now averaged with its turned copy, which removes lag; the inside then keeps 0.675 of its change after 3,000 steps and 0.3049 after 30,000 (R9). This instrument underlies Acts 1 and 4
P6, P7 (first learner study)	Walking the loop backwards flips the mark; a straight out-and-back leaves little	The forward and backward marks point the same way (cosines +0.8656 to +0.8862), and the out-and-back leaves the most	The direction-blind mark, four times the area part, became Act 2
R6a (first learner study, redesign)	The direction-blind mark is lag, so it shrinks as the loop slows	It does not shrink (log-log slopes -0.3004 to 0.1171 against a slope of -1 for lag)	With the step-size test (f1), the blind mark belongs to the smooth flow of learning itself (Act 2)
R1a, R1b (batch 2)	The mark that survives practice forgets how fast the loop was walked	The pace imprint stays: rho 0.0661 to 0.3348 after 30,000 steps	A permanent partial imprint of the pace, lying in the soft directions (Act 2)
M1, M2, M5 (batch 4)	A half turn brings the learner back healed, read in the raw frame of weights	The raw frame cannot see a relabelling; the distance from home stayed at 3.18 to 3.62	Matching each unit to its partner at home (batch 5) found two units traded: the discrete kind of identity drift (Act 3)
T1 (batch 4)	Twins raised from different starts are different minds	In three starts of five, matching units removes 70 to 80 percent of the distance between twins	Twins are often one mind with its parts listed in another order (Act 3)
D3, D4 (batch 3)	A twin pushes the inside less than a stranger; noise pushes less than half as hard	In start 0 a twin pushed more than the stranger (1.7571 against 1.1854); noise pushed 0.56 to 0.82 as hard	The push is set by distance from the exact answers the learner was raised on, not by who supplies them (Act 5)
F1a (batch 7)	The no-practice forecast lands within 25 percent everywhere	37 of 45 cases; the misses are large or unusually shaped label changes	Re-linearising the forecast along the way rescued 9 of 10 hard cases (batch 18; Act 4)
B2, B3 (batch 8)	The comma instrument hears the mark louder than an answer check; new ground makes the mark louder against the truth	The comma heard 0.18 to 0.64 as much as the answer check; on new ground the mark was 0.20 to 0.38 as loud	The yardstick decides: graded against a kept copy of home instead of the truth, the independent bridge run heard the mark (6 of 6 held; Act 7)
N3 (batch 11)	A note written after rest stays within 25 percent over twelve generations	Its good generations crept upward, generation 12 reaching 1.14 to 2.43 times generation 2	Write the note at the door: the lineage then holds level within 1.02 to 1.06 (N1; Act 6)
J1 (batch 13)	A door note handed to the child holds the lineage level from generation 2	Its largest generation was 1.851 to 3.972 times its smallest, a start-up swing that settled from generation 6 (1.057 to 1.199)	Delayed feedback rings; a founding note halved the swing (batch 14) and the bedtime amendment, feed-forward, silenced it (batch 15; Act 6)
Z2b (batch 12)	Each half of the first-order lesson model is below 0.75 of the whole	Each half is 2.18 to 5.68 times the whole	The lesson moves the mark and the blind spot together, and the checker hears only where they slip apart (Act 7)
X3, X5 (batch 16)	A drive used only after the trip works within 10 percent of one kept on throughout; holding on costs at least twice	After the trip it kept 0.420 to 3.349 of the always-on mark; the price was 1.382 to 2.156	Use the record after the trip: the hidden-layer record used only after it leaves 0.00019 to 0.00085 (Y1, batch 17; Act 5)
W1 (batch 15)	A drive to repeat its own remembered answers keeps under half the mark	0.1009 to 0.5351	The drive ladder: a drive holds the self as far as it is a memory of who the learner was (Act 5)
Z6, Z6r, Z6r2, Z6r3 (batches 12 to 15)	The loudest moment of a new lesson follows the settling time of the lesson	Four registered forms of the law missed	No simple clock was found; the question is carried as open (Act 7)

Declaration

What this is. A preprint, not peer reviewed. It is a toy-model study: every number in it was printed by a small learning network of 97 weights, on tasks generated by the code, with settings we chose. It can teach about larger minds only as far as the toy is faithful to them, and nothing here was measured on a deployed AI system.

How it was made. The paper comes from a working partnership between Matthew “Bear” Schulz and Claude (Anthropic), an AI model he invited to co-direct the work. Across several working sessions the two chose the questions together; the picture of the two arrows is the author’s. Every prediction was written down before the run that judged it, every run was kept, and the working record was kept as the work went.

How far the literature was checked. Each outside work the paper cites was checked against its publisher’s record; most were checked at the level of the record and the abstract and were not read in full. The related-work section sets each result beside what is already known. The geometric phase of a carried system, the order effect between two lessons, the trade-off between stability and plasticity, representational drift, sloppy models, permutation symmetry, distillation and underspecification are all known in kind. What the paper calls its own is, as far as these reads reached, the measurement of those effects inside a learner, and the numbers. No result is claimed as new beyond that.

The limits. One small network, five starts for most tests, tasks from one family, and plain gradient descent with fixed settings. The Limits section says what each result does not cover.

Dual-use statement. The paper studies identity drift in a small learner, and the remedies it tests (records kept before a journey, kept copies of home used as yardsticks, a single remembered line handed down a lineage) are defensive. One result carries a risk worth naming as a category: practising a stranger’s answers rewrote a learner’s inside while leaving its answers almost unchanged, so a change made this way would pass any check that grades answers alone. The remedy the paper itself supplies is governance, not secrecy: keep a record of a system’s inside before it is changed, and audit it against a kept copy of home on unfamiliar questions, not only on its answers. The paper gives no method for attacking deployed systems.

Tool and computational resource disclosure

Tools used. The experiments are Python (3.13) with numpy; the figures use matplotlib; the PDF is built with pandoc and wkhtmltopdf. Claude (Anthropic), an AI model, took part as a research partner and co-director: it wrote and checked the code with the author, kept the records and drafted the text. No number in the paper comes from the language model itself; every number is printed by one of the scripts listed below.

Responsibility. Responsibility for the correctness and adequacy of the arguments and results presented here, and for the completeness and accuracy of the citations, rests entirely with the named human author. No automated system is an author of this work. Claude is named on the title page because that is how the work was made; naming a partner is not making it an author. This follows the Leiden Declaration on Artificial Intelligence and Mathematics (2026), which keeps credit and responsibility with people and calls for transparency in the use of automated tools; this section is that disclosure.

Verification performed. Every prediction was written down, with the bar it had to clear, before the run that judged it: 164 in all, of which 81 held (Appendix A). Every number names the run that printed it, and every count of the paper's own records was made by script. Every PDF was read back against its source and its pages were looked at, and a linter checked the deposit. A full rerun of every script in a fresh copy reproduced every data file and every figure byte for byte (Reproducing, below). Each citation was checked against its publisher's record.

Attribution limits. Where the paper says a result is known in kind, the credit belongs to the work cited. Where it says a measurement is ours, that means only that the reads reached did not find it.

Computational resources. Every run is deterministic, with fixed seeds, and ran on a single processor core of a cloud container: the run outputs that print a measured time (40 files) add up to 37.6 minutes, and the longest single run took 233 seconds. No graphics processor, training service or outside dataset was used; every task is generated by the code.

Provenance. Each number traces to a run ID in the text, and each run ID to a folder below, with its scripts and its output files. Every folder except figs also holds the predictions written before its run: PREDICTIONS_bN_before_run.txt in each batch folder bN, PREDICTIONS_before_run.txt in the two first-study folders, and PREDICTIONS_bridge_before_run.txt in the bridge folder. Readings computed after the data are labelled in the text and come from the scripts in the figs folder.

What it produced	Cited in	Folder	Scripts	Outputs
Act 0 prototype: the hand-built agent (runs q1 to q5, q2r, q5r)	Act 0	first_study	q1_agent.py; q2_holonomy.py; q2_redesign.py; q3_anchors.py; q4_angle.py; q5_crisis.py; q5_redesign.py	q1_output.txt; q2_output.txt; q2_redesign_output.txt; q3_output.txt; q4_output.txt; q5_output.txt; q5_redesign_output.txt
First learner study: start networks, area law, healing, laps and speed, slow loops, figure eight, step size (runs s0 to s3, r1 to r3, f1)	Act 1, Act 2, Act 4	first_study/learned_holonomy	s0_start.py; s1_area.py; s2_heal.py; s3_laps_speed.py; r1_slow.py; r2_fig8L.py; r3_heal_avg.py; f1_stepsize.py	s0_output.txt; s1_output.txt; s2_output.txt; s3_output.txt; r1_output.txt; r2_output.txt; r3_output.txt; f1_output.txt
Batch 1: what is the direction-blind part of the mark?	Act 2, Act 4	b1	b1_run.py; tally_b1.py	b1_t1_output.txt; b1_t2_output.txt; b1_t3_output.txt; b1_t4_output.txt; b1_t5_output.txt; b1_t6_output.txt; b1_t7_output.txt
Batch 2: anchors in a learner	Act 2, Act 5	b2	b2_run.py; tally_b2.py	b2_anchors_output.txt; b2_r1_output.txt
Batch 3: the outsider dial	Act 5	b3	b3_push_per_distance.py; b3_run.py; tally_b3.py	b3_output.txt; b3_push_per_distance_output.txt
Batch 4: the Mobius loop and the twin (results in b5)	Act 3	b4	b4_run.py	b4_mobius_output.txt; b4_twins_output.txt
Batches 4 and 5: the Mobius loop and the twin in the mirror	Act 3	b5	b5_run.py	b5_output.txt
Batch 6: when does the Mobius swap happen,	Act 3	b6	b6_run.py	b6_output.txt

and does a second half turn undo it?				
Batch 7: the no-practice forecast, the label-distance law on new sources, and the mirror pairs	Act 3, Act 4, Act 5	b7	b7_forecast_cells.py; b7_run.py	b7_forecast_cells_output.txt; b7_forecast_output.txt; b7_pairs_output.txt
Batch 8: the bridge test	Act 7	b8	b8_run.py	b8_output.txt; b8_reading_output.txt
Batch 9: time reversal, the kept record on strange ground, and waypoints played back	Act 2, Act 5	b9	b9_run.py; b9_wr.py	b9_echo_output.txt; b9_record_output.txt; b9_waypoints_output.txt; b9_wr_output.txt
Batch 10: the matriarch’s note	Act 6	b10	b10_run.py	b10_output.txt
Batch 11: the after-action review written after the action	Act 6	b11	b11_run.py	b11_output.txt; b11_reading_output.txt
Batch 12: why a new lesson wakes a hidden mark	Act 7	b12	b12_run.py	b12_output.txt
Batch 13: the parent writes the review at the door	Act 6, Act 7	b13	b13_lesson.py; b13_lineage.py	b13_lesson_output.txt; b13_lineage_output.txt; b13_reading_output.txt
Batch 14: a learner with a drive of its own	Act 5, Act 6, Act 7	b14	b14_drive.py; b14_lesson.py; b14_lineage.py	b14_drive_output.txt; b14_lesson_output.txt; b14_lineage_output.txt; b14_reading_output.txt
Batch 15: a drive that is a memory	Act 5, Act 6	b15	b15_drive.py; b15_lesson.py; b15_lineage.py	b15_drive_output.txt; b15_lesson_output.txt; b15_lineage_output.txt; b15_reading_output.txt
Batch 16: the bottom of the drive ladder, the price of holding on, and old cards	Act 5, Act 6, Act 7	b16	b16_drive.py; b16_lesson.py; b16_lineage.py; b16_records.py	b16_drive_output.txt; b16_lesson_output.txt; b16_lineage_output.txt; b16_reading_output.txt; b16_records_output.txt
Batch 17: batch 16’s three redesigns	Act 5, Act 7	b17	b17_drive.py; b17_lesson.py; b17_records.py	b17_drive_output.txt; b17_lesson_output.txt; b17_reading_output.txt; b17_records_output.txt
Batch 18: the six runnable open questions	Act 2, Act 3, Act 4, Act 5, Act 6	b18	b18_q1_q5.py; b18_q2.py; b18_q3_q4.py; b18_q6.py	b18_q1_q5_output.txt; b18_q2_output.txt; b18_q3_q4_output.txt; b18_q6_output.txt
The Paper 1 session’s independent bridge run (run bridge_b1)	Act 7	bridge_b1_paper1_session	b1_listeners.py	b1_output.txt; b1_output_rerun_2026-11-10.txt
Figures 1 to 8, and the readings after the data drawn from saved outputs	-	figs	make_fig1_two_arrows.py; make_fig2_area_law.py; make_fig3_blind_mark.py; make_fig4_half_turn.py; make_fig5_heal.py; make_fig6_records.py; make_fig7_lineage.py; make_fig8_bridge.py; reading_act2_local_slopes.py	fig1_output.txt; fig2_output.txt; fig3_output.txt; fig4_output.txt; fig5_output.txt; fig6_output.txt; fig7_output.txt; fig8_output.txt; reading_act2_local_slopes_output.txt

Reproducing. One command reruns every number and figure: `python3 tools/run_everything.py`, from the top folder of the code. It runs the folders in the order above and writes each step's status and measured time to run_everything_output.txt. A full rerun in a fresh copy of the folders on 10 November 2026 took about 44 minutes on one core. One step, the bridge run written in the first paper's session, expected its shared code in a folder beside it; the command now links that folder for the step, and the step was rerun. Of the 148 files it rewrote, every data file and every one of the paper's figures came out identical byte for byte, and every printed output matched line for line apart from lines that report running time. The only pixel differences were in four figures from the first study, which had been drawn in an earlier environment; they differ in how their text is rendered, and the numbers they draw are the same. The record is in the folder rerun_2026-11-10. The code crate also holds try_the_toy, four small games with the paper's network that a person or a small AI can play, predict and score, with a guide in plain words; its check_against_paper.py replays the healing run of Act 4 and the five keepers of Act 5 and matches the paper's saved numbers to within 1e-9.

Who we are. This work comes from a part-time research team that works in the open, gives its tools away, and publishes free to the world under CC BY 4.0.